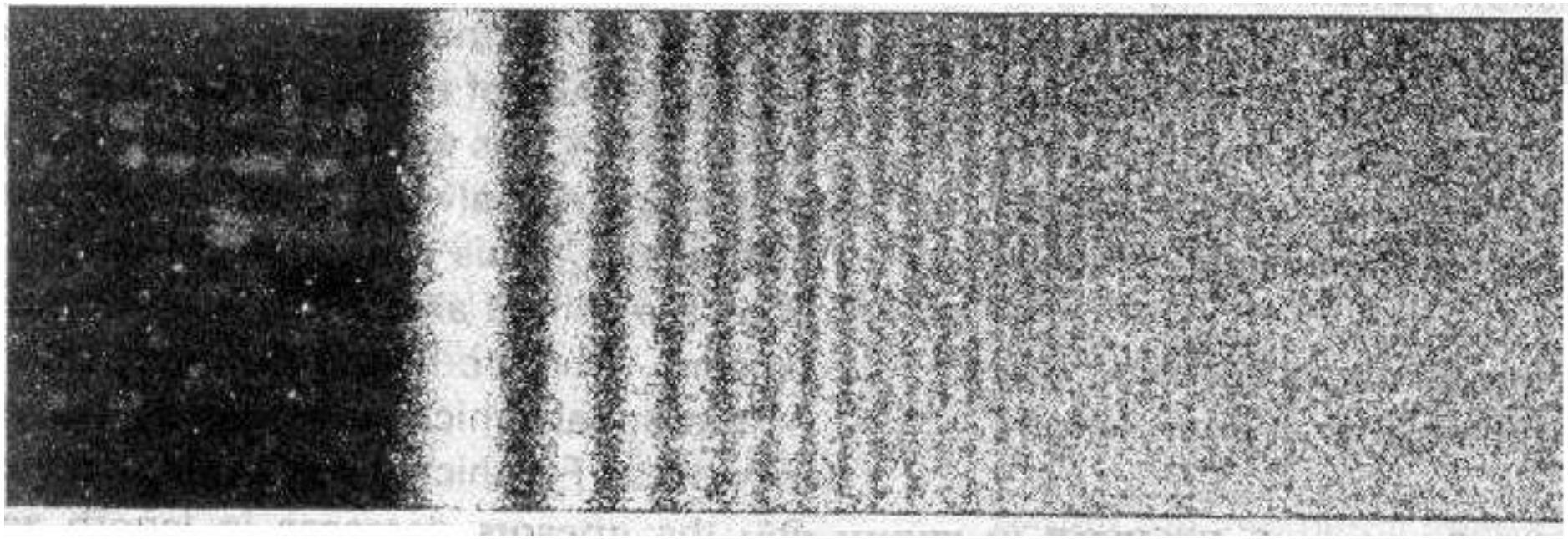
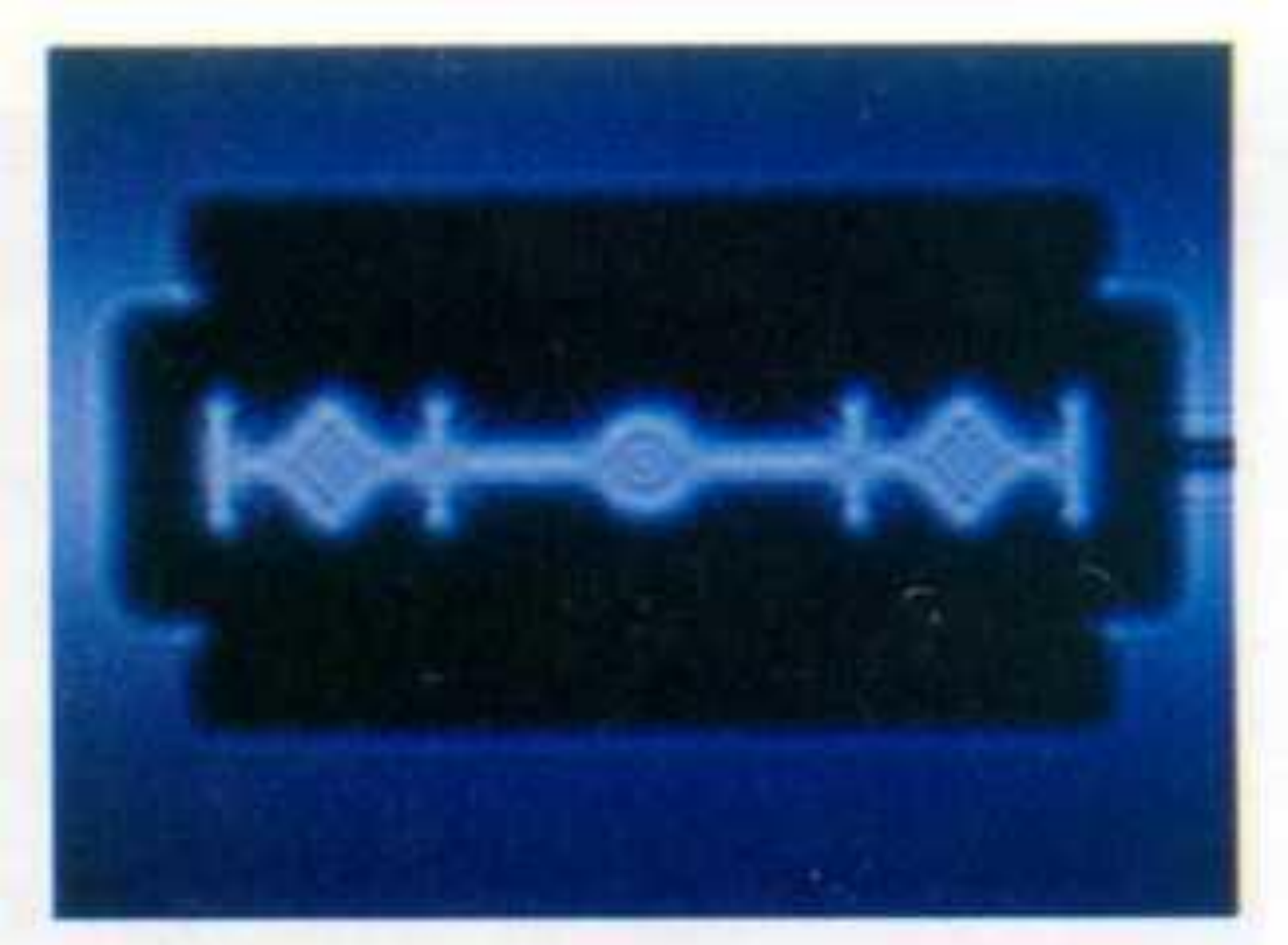
光的衍射（Diffraction of light）

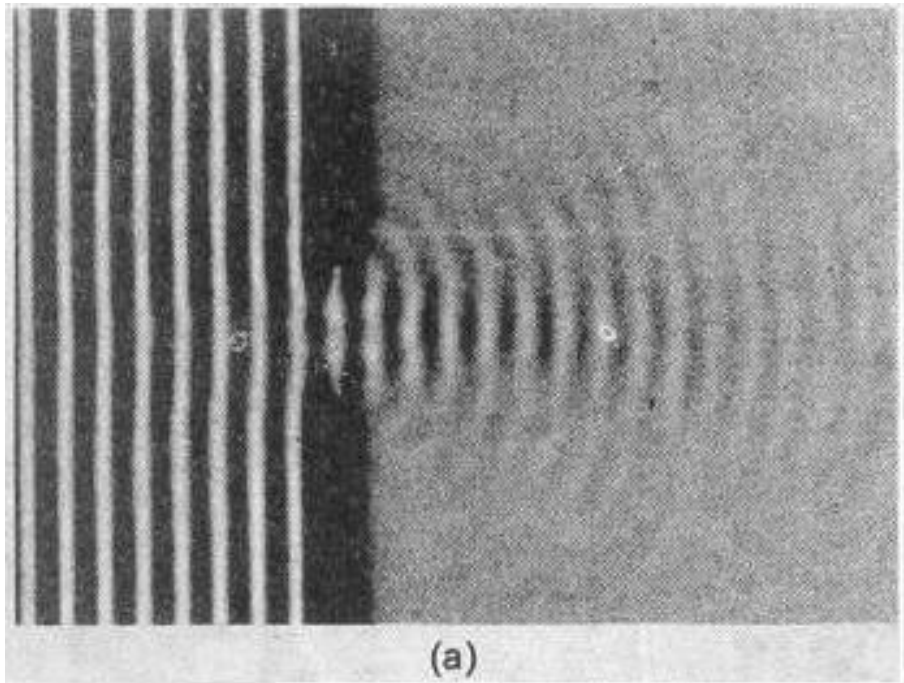


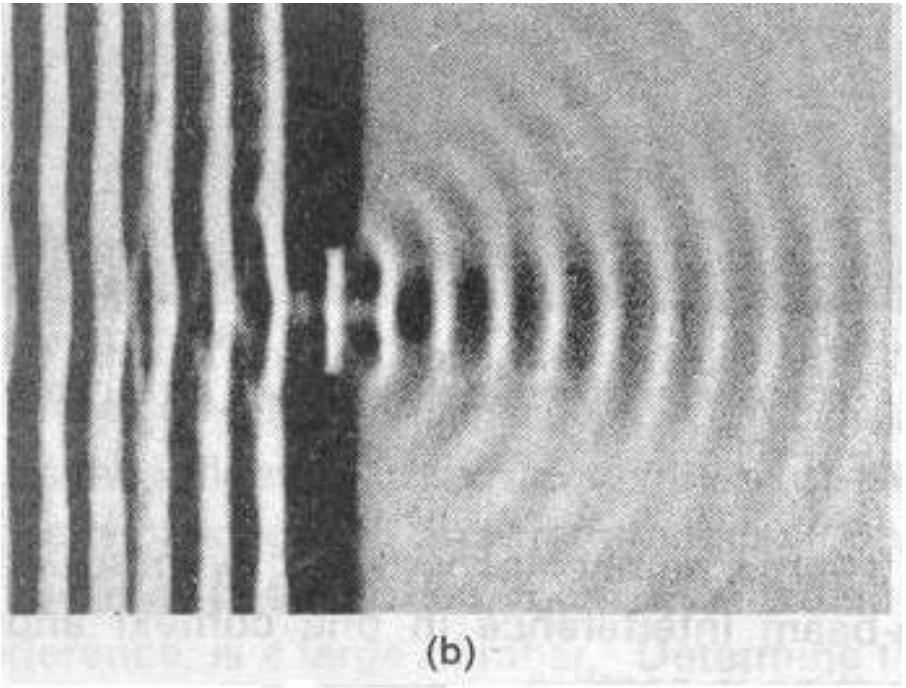
半屏边缘的衍射图案

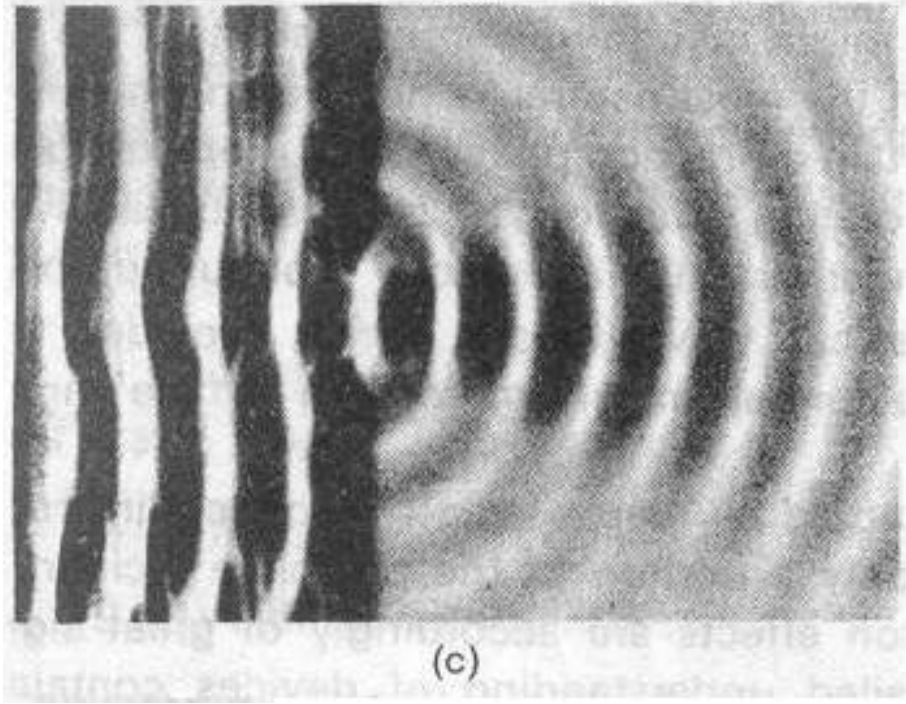
光在障碍物的边缘偏离直线传播 — 衍射现象



刀片周围衍射图案







水波通过小孔的衍射

认识了惠更斯原理

光的衍射（Diffraction of light）

§1　衍射现象、惠更斯——菲涅耳原理

§2　单缝的夫琅禾费衍射、半波带法

§3　光栅衍射

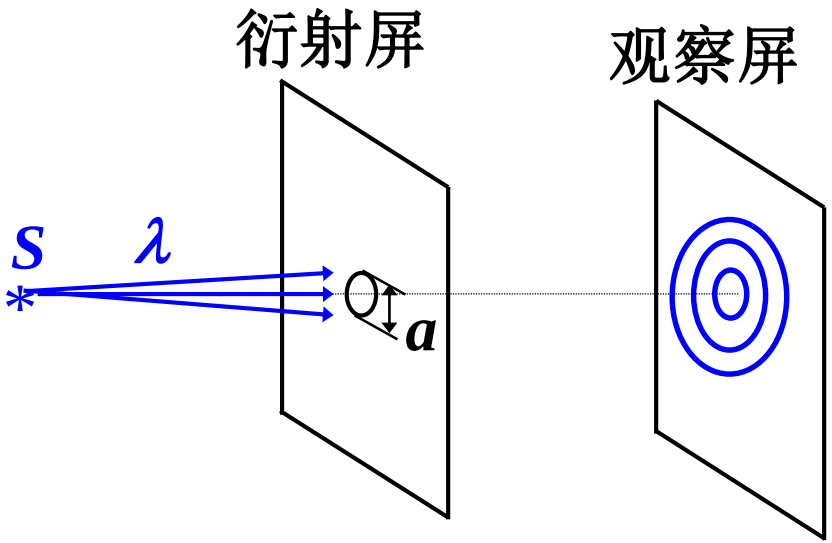
§4　光学仪器的分辨本领

§5　X射线的衍射

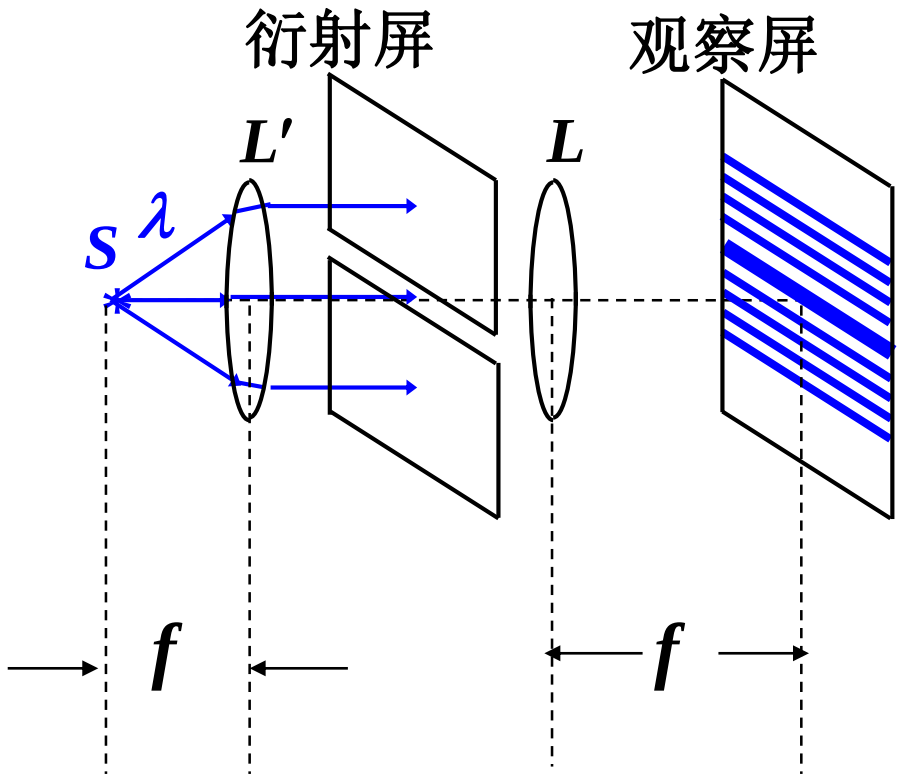
§1　衍射现象、惠更斯——菲涅耳原理

一、光的衍射

1. 现象:



$\lambda \geq 10^{-2} a$



3. 分类：

(1) 菲涅耳衍射　　近场衍射

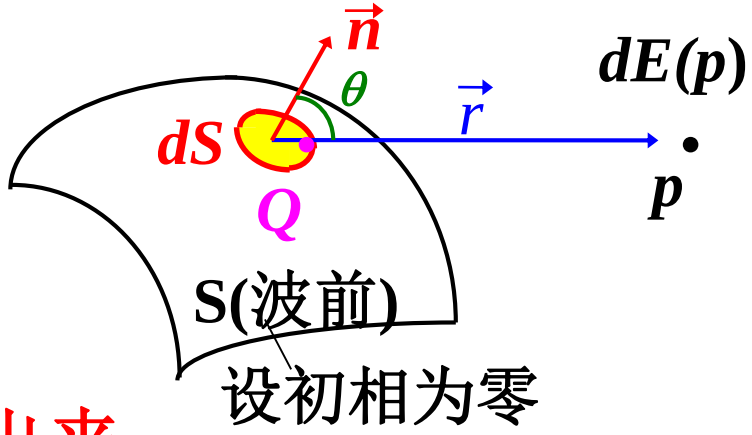
(2) 夫琅禾费衍射　远场衍射

光源和观察屏（或二者之一）离衍射孔（或缝）的距离有限

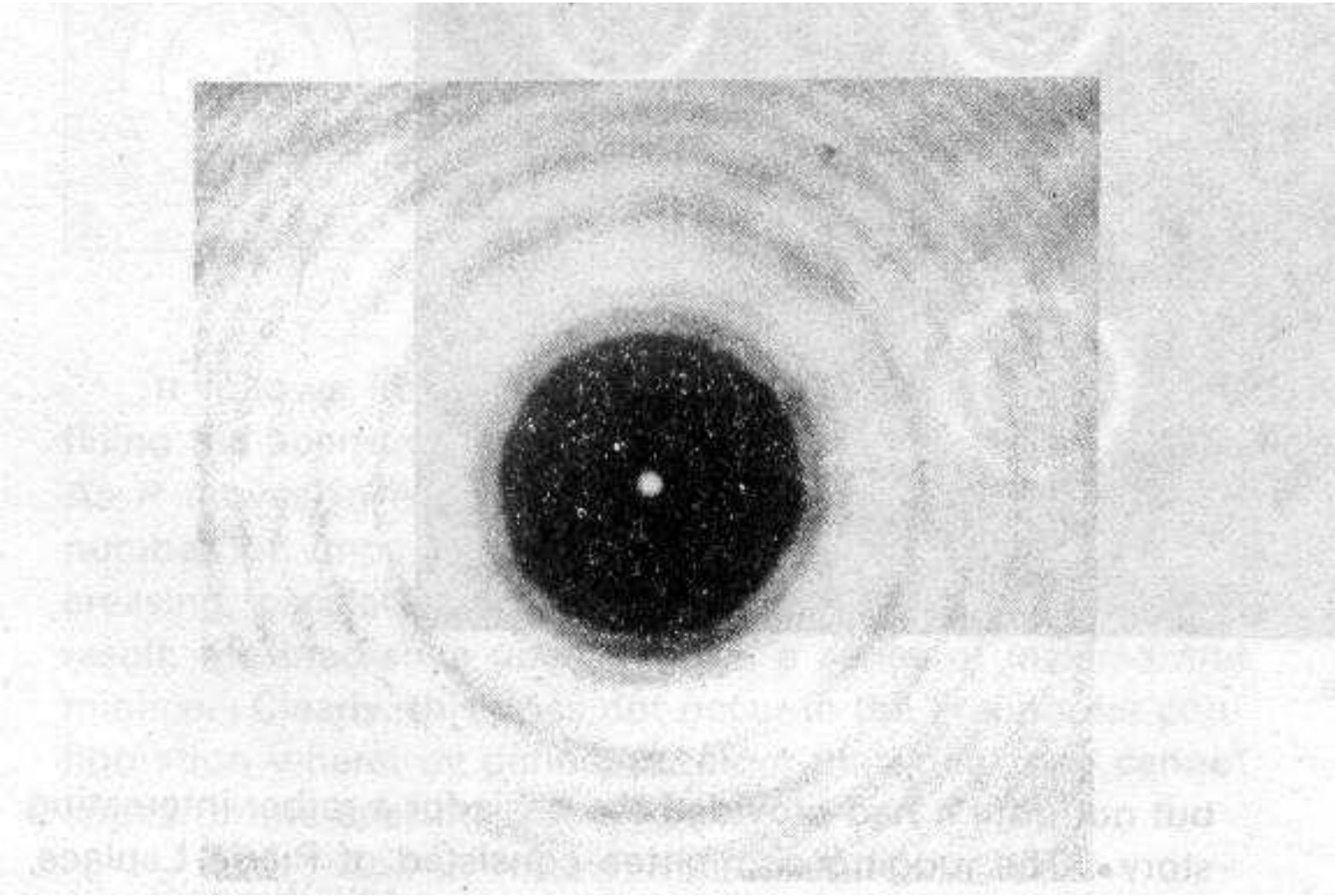
二、惠更斯——菲涅耳原理

波传到的任何一点都是子波的波源，各子波在空间某点的相干叠加，就决定了该点波的强度。

$$dE_{(p)} \propto \frac{a(Q)K(\theta)}{r} dS$$



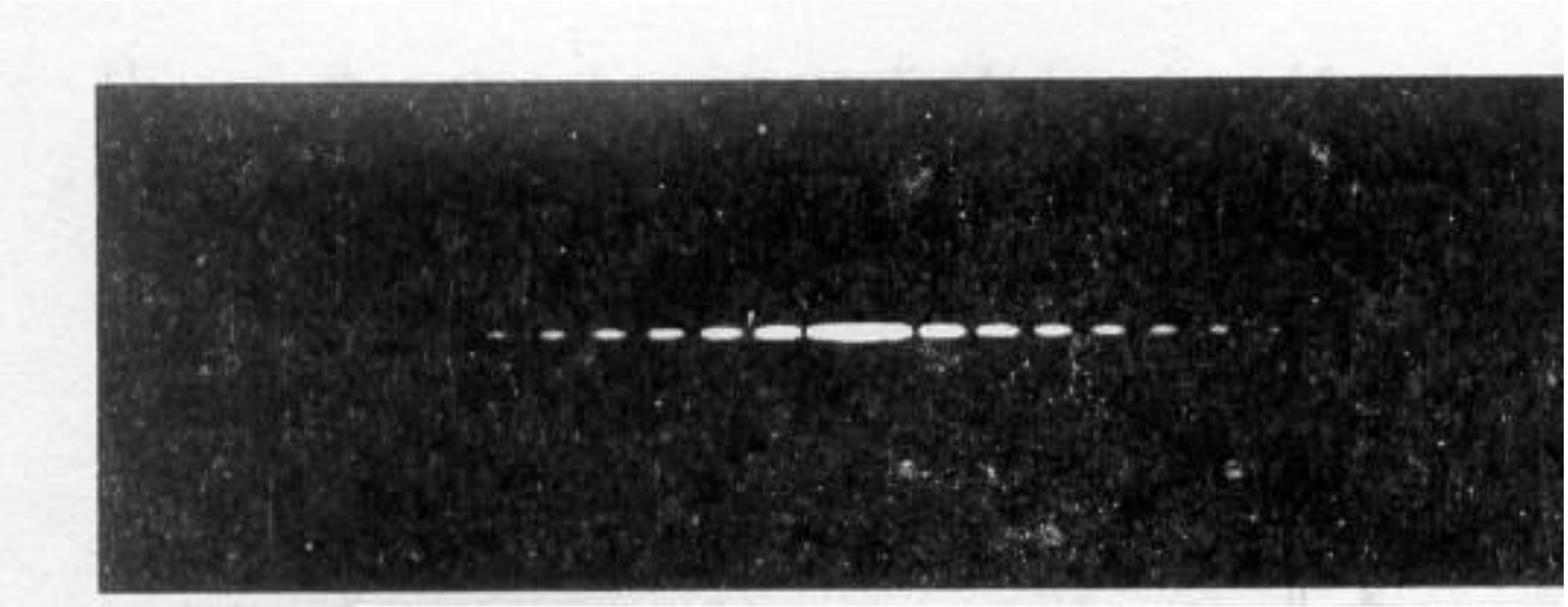
利用麦克斯韦方程组可以推导出来



泊松亮点

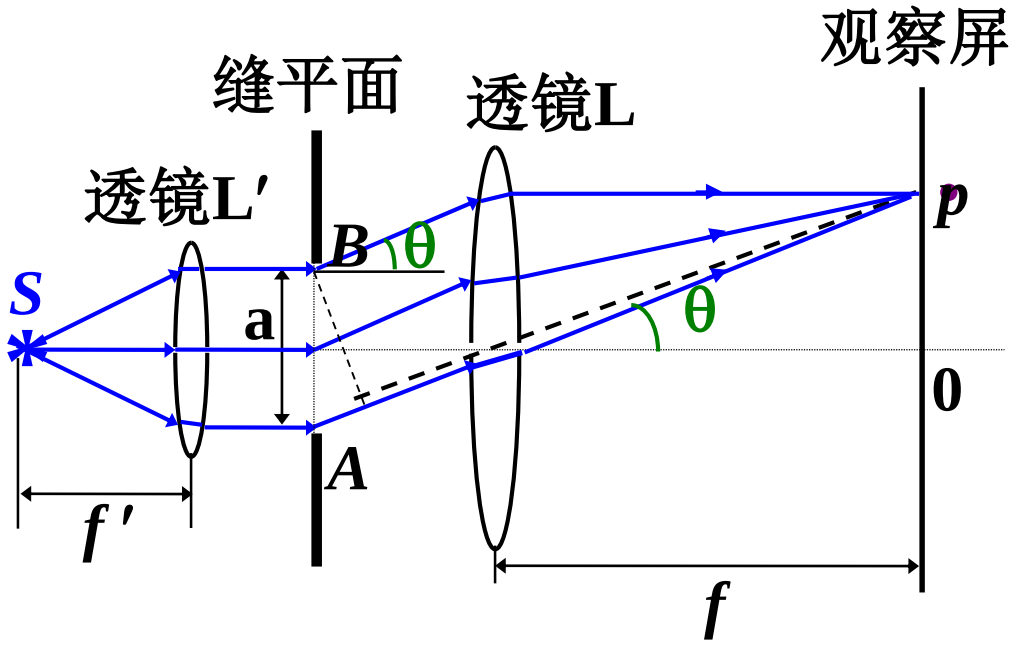
小球后面中心是亮的

§2 单缝的夫琅禾费衍射、半波带法



点光源照射单缝（竖直）时的衍射图案

一. 装置



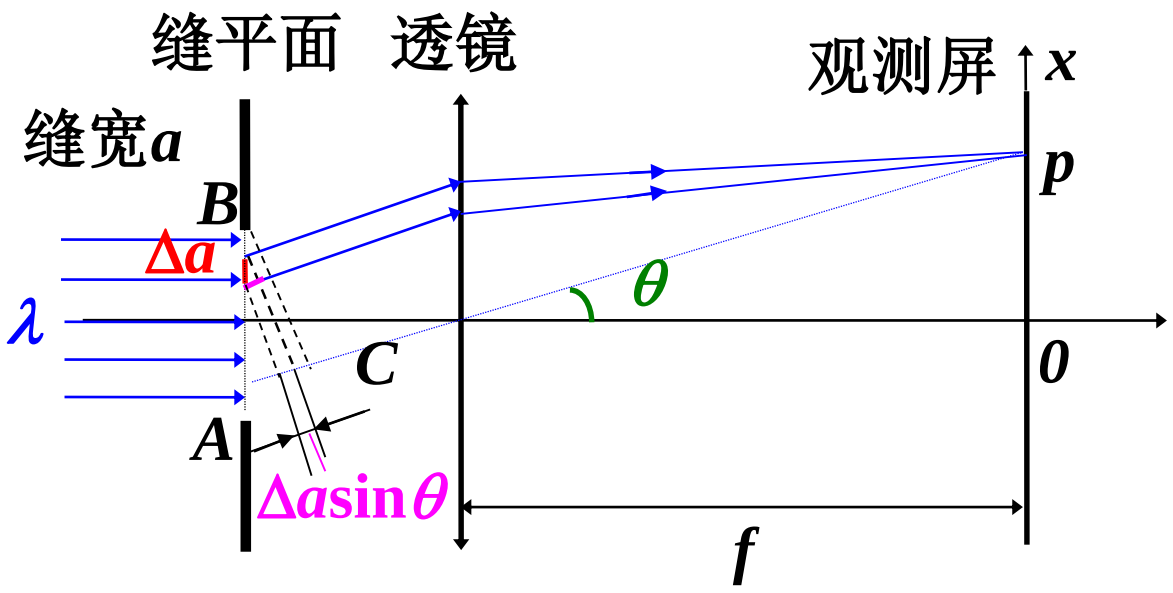
S：单色相干光源

θ ：衍射角

$\overline{AB} = a$（缝宽）

p点光强是缝作为子波源，
在 p处相干叠加的结果

沿狭缝方向光强分布与光源大小有关



子波的
相干叠加

- $A \rightarrow P$ 和 $B \rightarrow P$ 的光程差

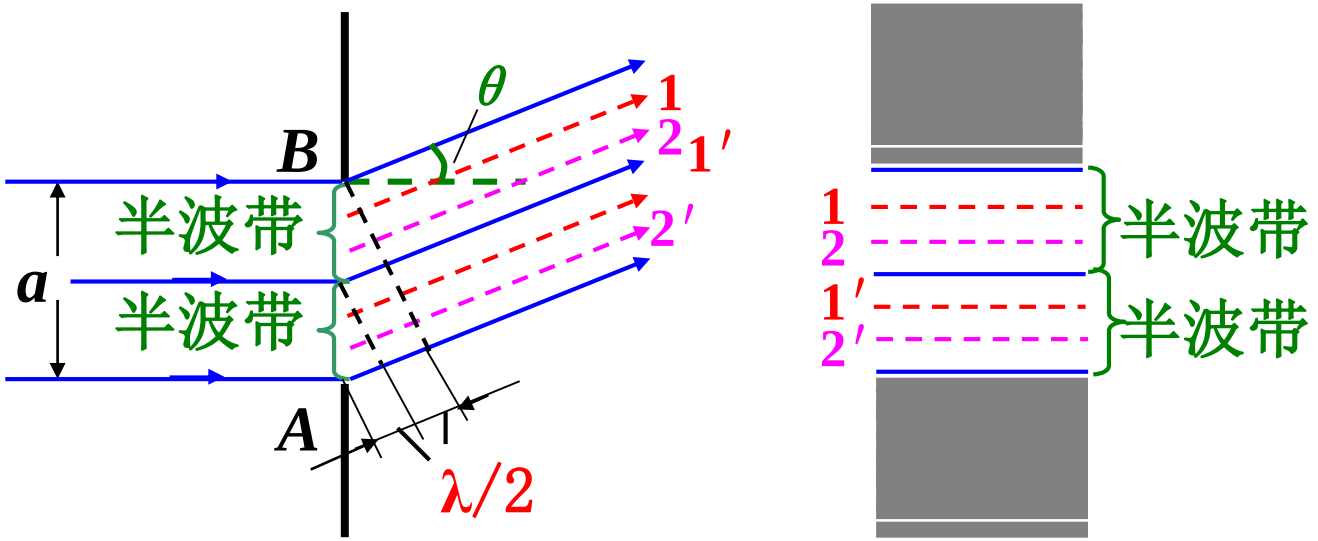
$$\delta = a\sin\theta \quad = \lambda$$ 是亮纹还是暗纹?

二. 半波带法

衍射主极大

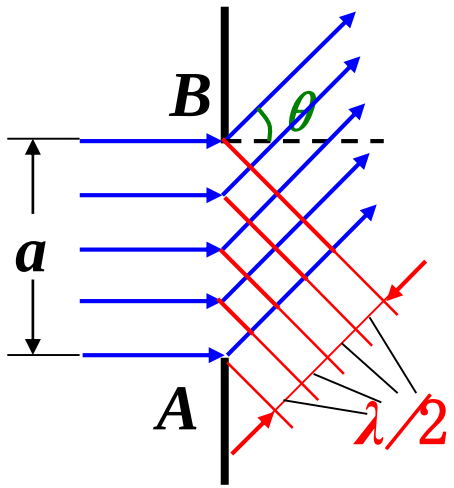
$\theta = 0,\ \ \delta = 0$ —— 中央明纹(中心)

- 当 $a\sin\theta=\lambda$ 时，可将缝分为两个“半波带”



两个“半波带”的光在*P*处干涉相消形成暗纹

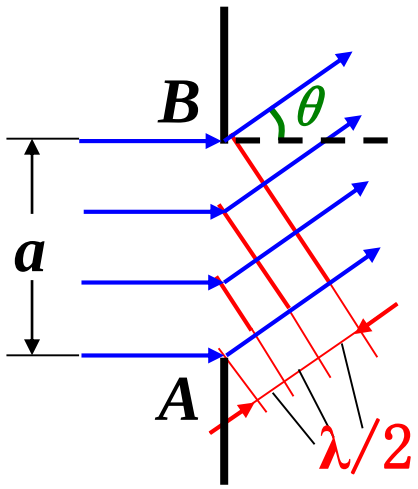
- 当 $a\sin\theta=2\lambda$ 时，
可将缝分成四个“半波带”



形成暗纹

一般情况 $a\ \sin\theta = \pm k\lambda,\ \ k = 1, 2, 3\cdots$ ——暗纹

- 当 $a\sin\theta = \dfrac{3}{2}\lambda$ 时，
可将缝分成三个“波带”



P处近似为明纹中心

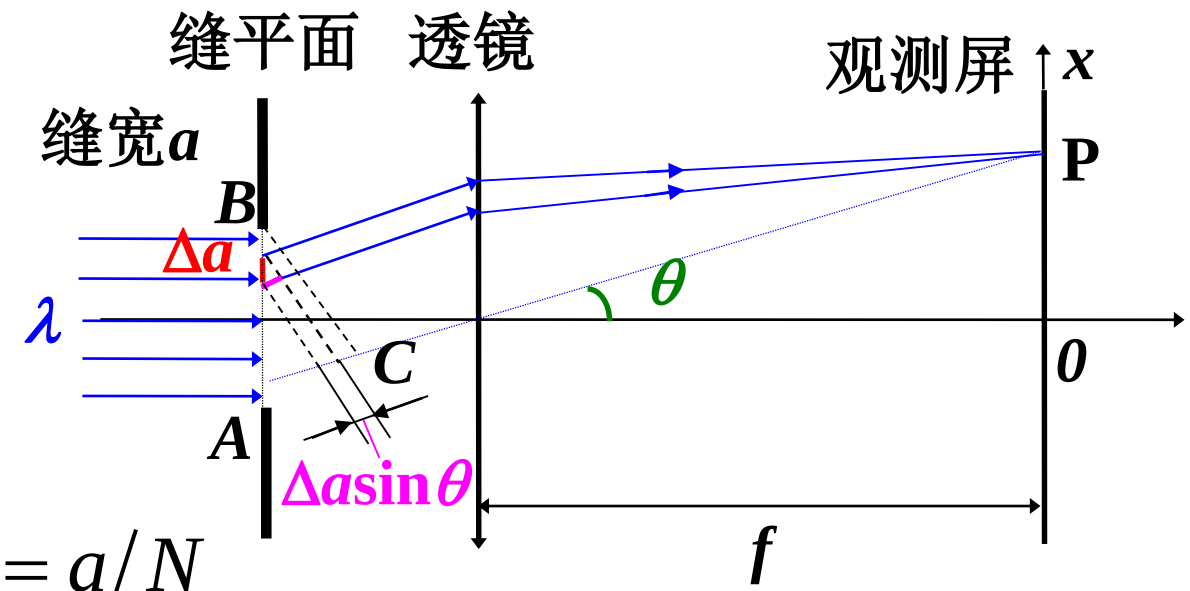
$$a\sin\theta = \pm(2k'+1)\frac{\lambda}{2},\quad k' = 1, 2, 3\cdots$$

——明纹(中心)

$a\sin\theta = 0$ ——中央明纹(中心)

上述暗纹和中央明纹(中心)位置是准确的，
其余明纹中心的位置较上稍有偏离。

三. 振幅叠加、光强公式

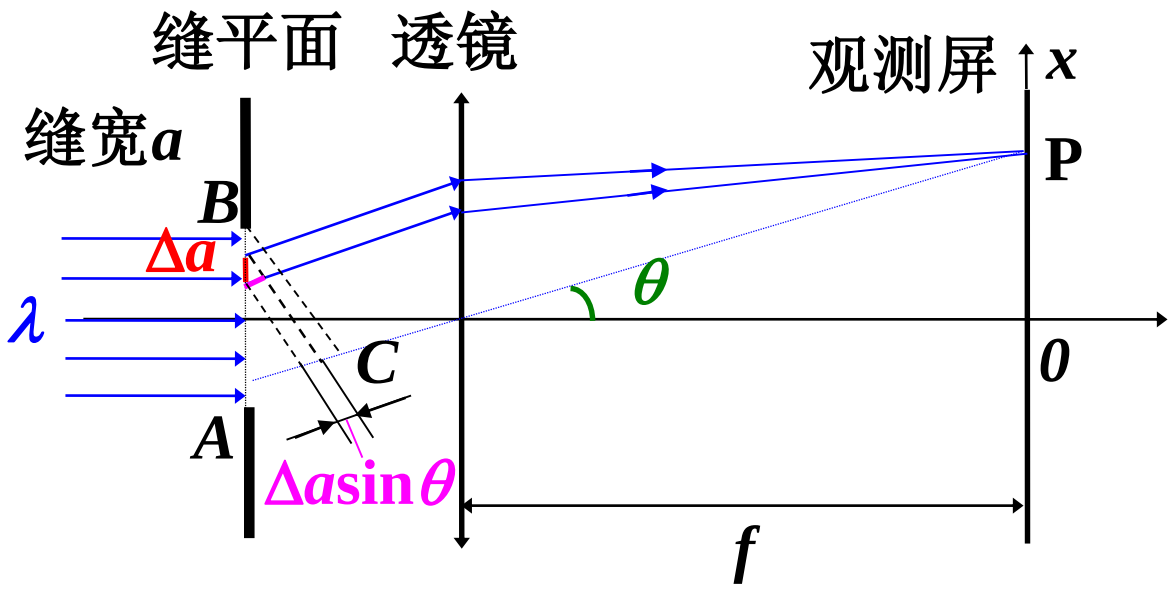


$$\Delta a = a/N$$

每个窄带发的子波在P点振幅近似相等，设为 ΔE_0

$$\Delta\varphi = \frac{\Delta a \sin\theta}{\lambda}\cdot 2\pi = \frac{a\cdot\sin\theta}{N}\cdot\frac{2\pi}{\lambda}$$

各子波是相干波源



$$E_P = \Delta E_0 \sum_{m=1}^{N} \cos[\omega t + (m-1)\Delta\varphi]$$

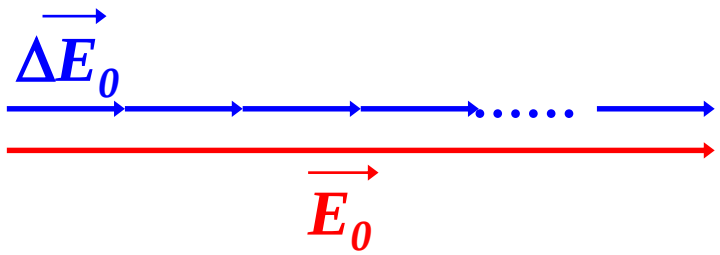
代数：利用复数

$$E_P = \Delta E_0 \,\mathrm{Re}\left\{ \sum_{m=1}^{N} \mathrm{e}^{i[\omega t + (m-1)\Delta\varphi]} \right\}$$

矢量法：

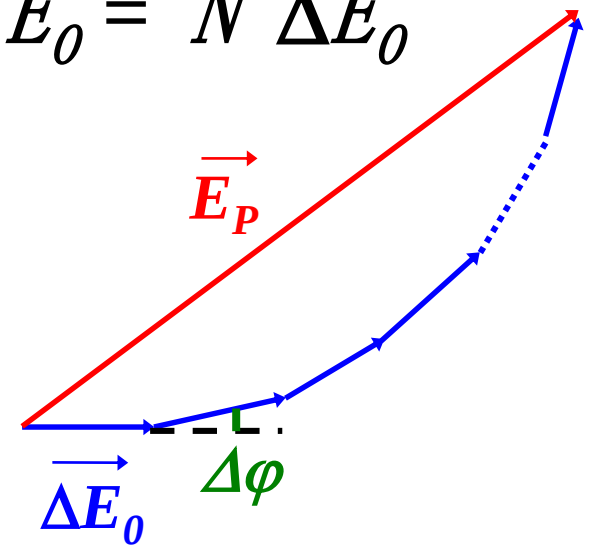
$$E_P = \Delta E_0 \sum_{m=1}^{N} \cos[\omega t + (m-1)\Delta\varphi]$$

对于0点：$\theta = 0$，$\Delta\varphi = 0$



$E_0 = N\,\Delta E_0$

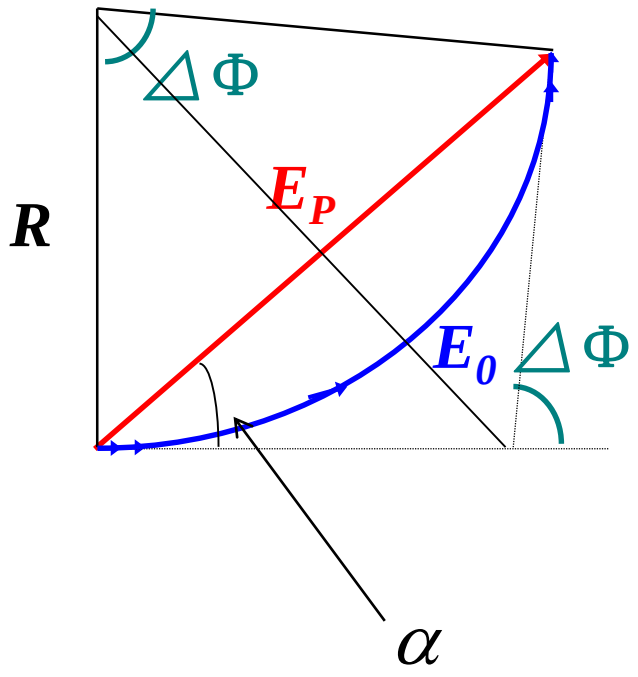
对于其他点P：$E_P < E_0$



当$N \to \infty$时，N个相接的折线将变为一个圆弧。

$$\Delta\Phi = N\Delta\varphi = \frac{a\sin\theta}{\lambda}2\pi$$

$$E_p = 2R\sin\frac{\Delta\Phi}{2}, \qquad E_0 = R\Delta\Phi$$



$$E_p = 2\frac{E_0}{\Delta\Phi}\sin\frac{\Delta\Phi}{2} = \frac{E_0}{\Delta\Phi/2}\sin\frac{\Delta\Phi}{2}$$

令 $$\alpha = \frac{\Delta\Phi}{2} = \frac{\pi a\sin\theta}{\lambda}$$

$$E_p = E_0\frac{\sin\alpha}{\alpha}$$

P点的光强

$$I \propto E_p^2 \quad , \quad I_0 \propto E_0^2$$

$$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$$

分析衍射图案: $I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$

(1) 主极大（中央明纹中心）位置:

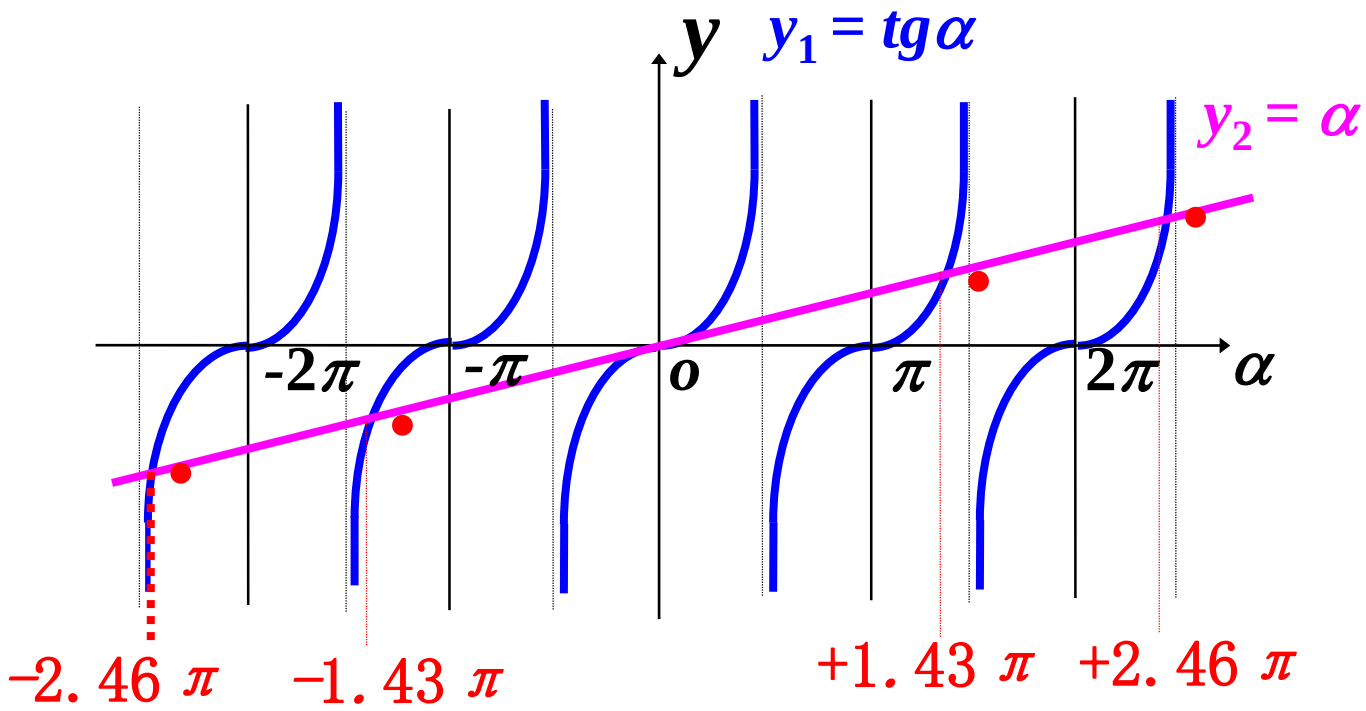
$\theta = 0$处，$\alpha = 0 \rightarrow \frac{\sin\alpha}{\alpha} = 1 \rightarrow I = I_0 = I_{\max}$

(2) 极小（暗纹）位置:

$\alpha = \pm k\pi$，$k = 1,2,3\cdots$时，$\sin\alpha = 0 \rightarrow I = 0$

由 $\alpha = \frac{\pi a \sin\theta}{\lambda} = \pm k\pi$ 得 $a\sin\theta = \pm k\lambda$

(3) 次极大位置: $\frac{dI}{d\alpha} = 0 \rightarrow tg\alpha = \alpha$

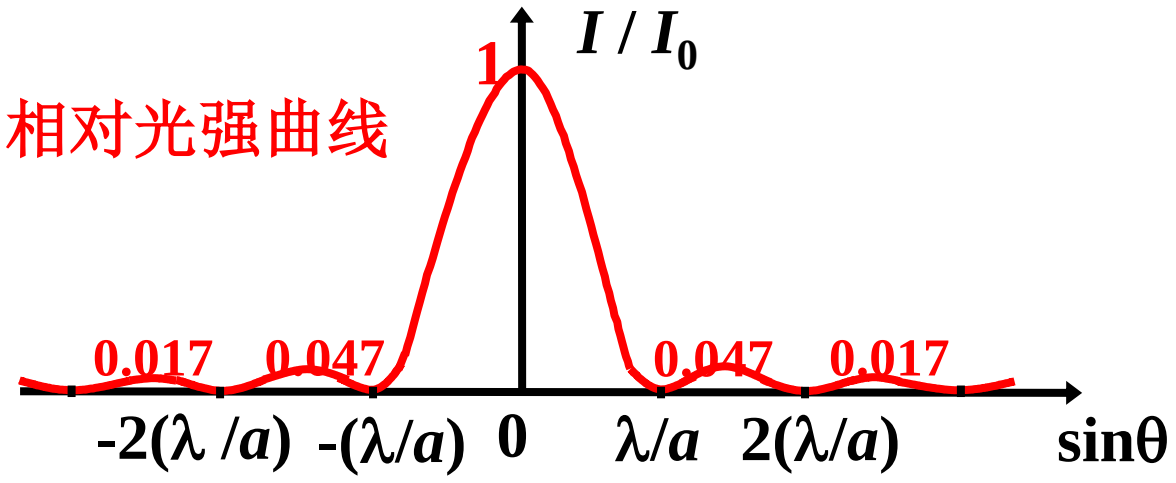


相应 ：$a\sin\theta = \pm 1.43\lambda, \pm 2.46\lambda, \pm 3.47\lambda, \cdots$

(4) 光强　从中央往外各次极大的光强依次为:

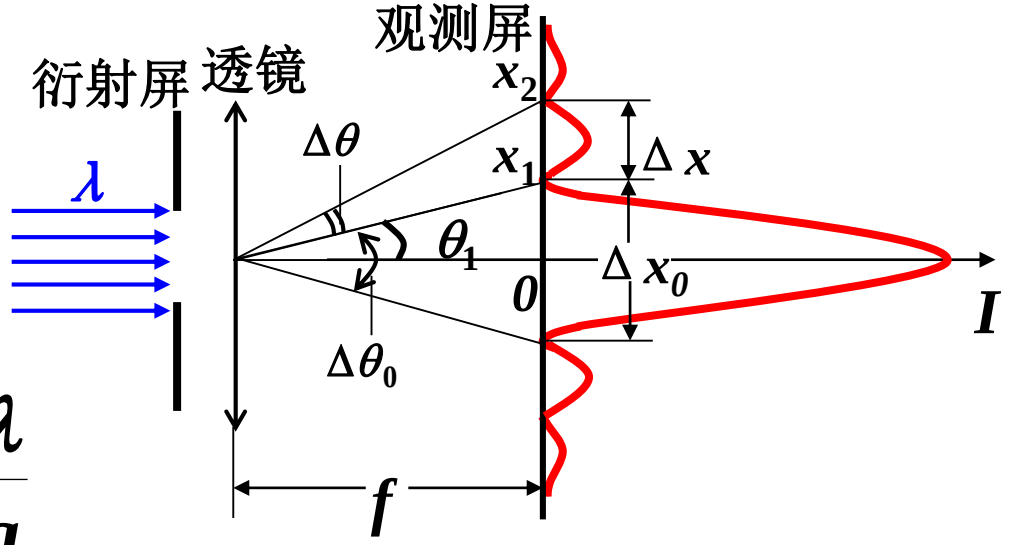
$$0.0472 I_0, \quad 0.0165 I_0, \quad 0.0083 I_0, \quad \cdots$$

$$\therefore \quad I_{次极大} \ll I_{主极大}$$



四. 条纹宽度

1. 中央明纹:

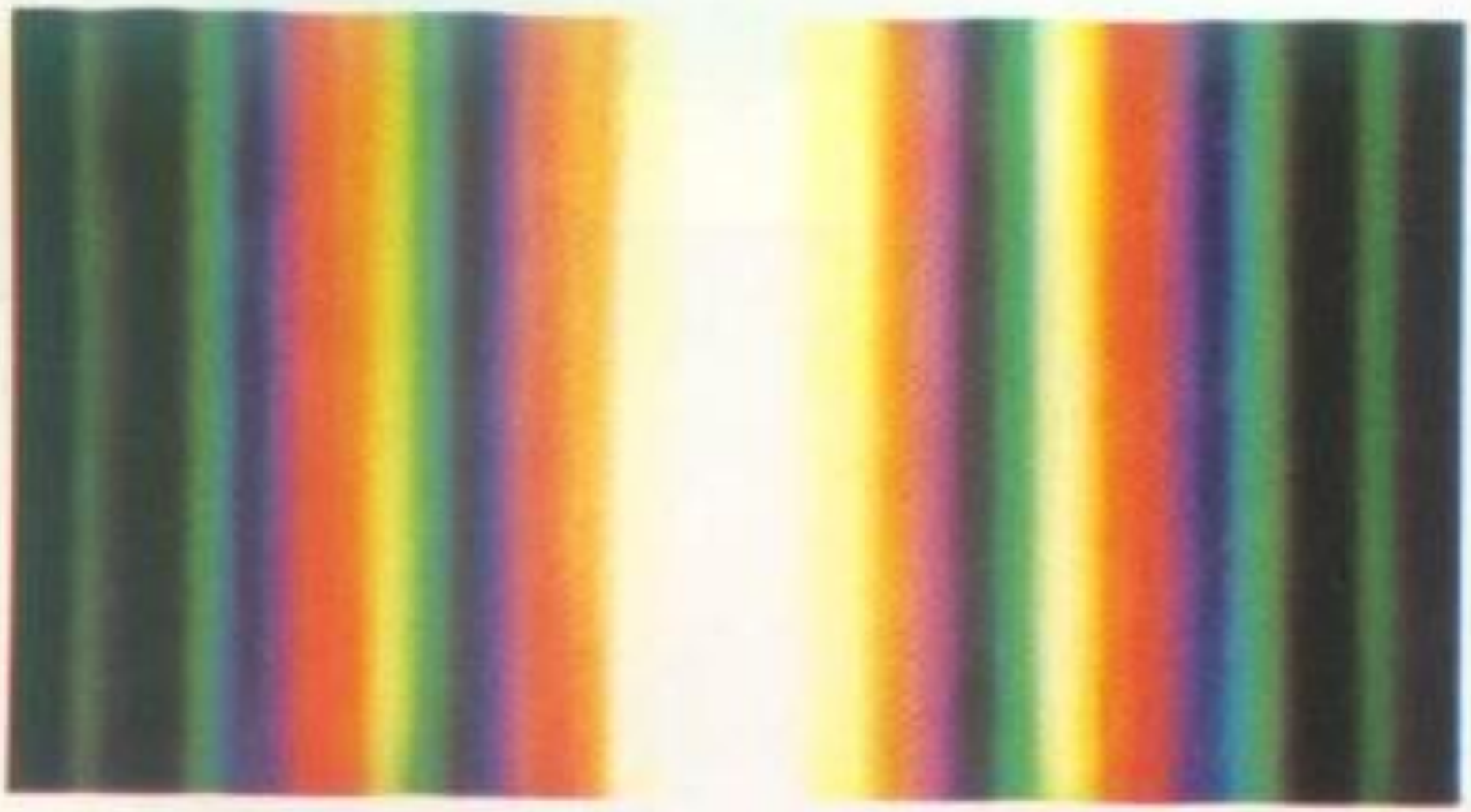


$a >> \lambda$ 时, $\sin\theta_1 \approx \theta_1$

角宽度 $\Delta\theta_0 = 2\theta_1 \approx 2\dfrac{\lambda}{a}$

线宽度 $\Delta x_0 = 2f \cdot tg\theta_1 = 2f\theta_1 = 2f\dfrac{\lambda}{a} \propto \dfrac{\lambda}{a}$

——衍射反比定律



Passing white light through a single slit produces this diffraction pattern.

2. 其他明纹(次极大)

$$\Delta x \approx \frac{f\lambda}{a} = \frac{1}{2}\Delta x_0$$

3. 波长对条纹宽度的影响

$\Delta x \propto \lambda$ 波长越长，条纹宽度越宽

4. 缝宽变化对条纹的影响

$\Delta x = \frac{1}{2}\Delta x_0 = f\frac{\lambda}{a}$ 缝宽越小，条纹宽度越宽

当 $\frac{\lambda}{a} \to 0$ 时， $\Delta x \to 0$

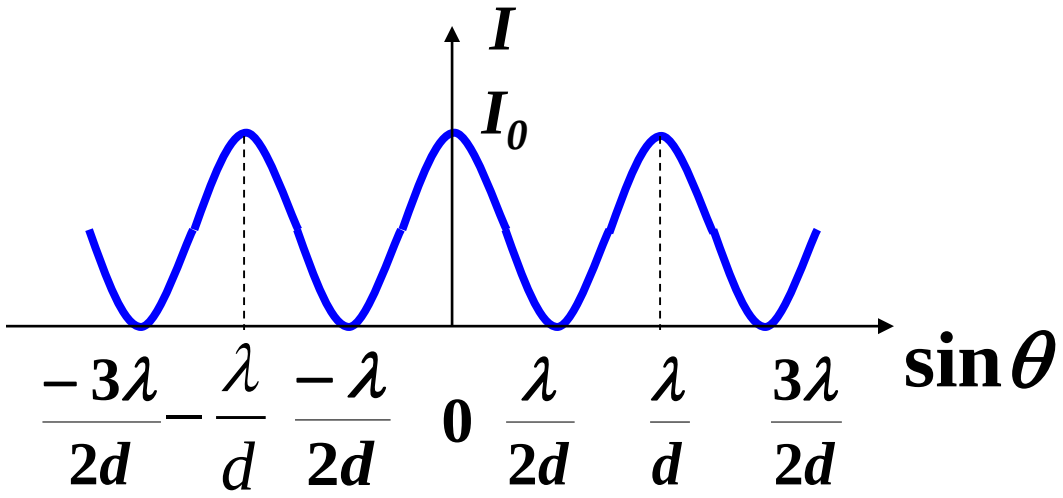
只显出单一的明条纹 ——单缝的几何光学像

∴几何光学是波动光学在$\lambda/a \to 0$时的极限情形

§3　光栅衍射

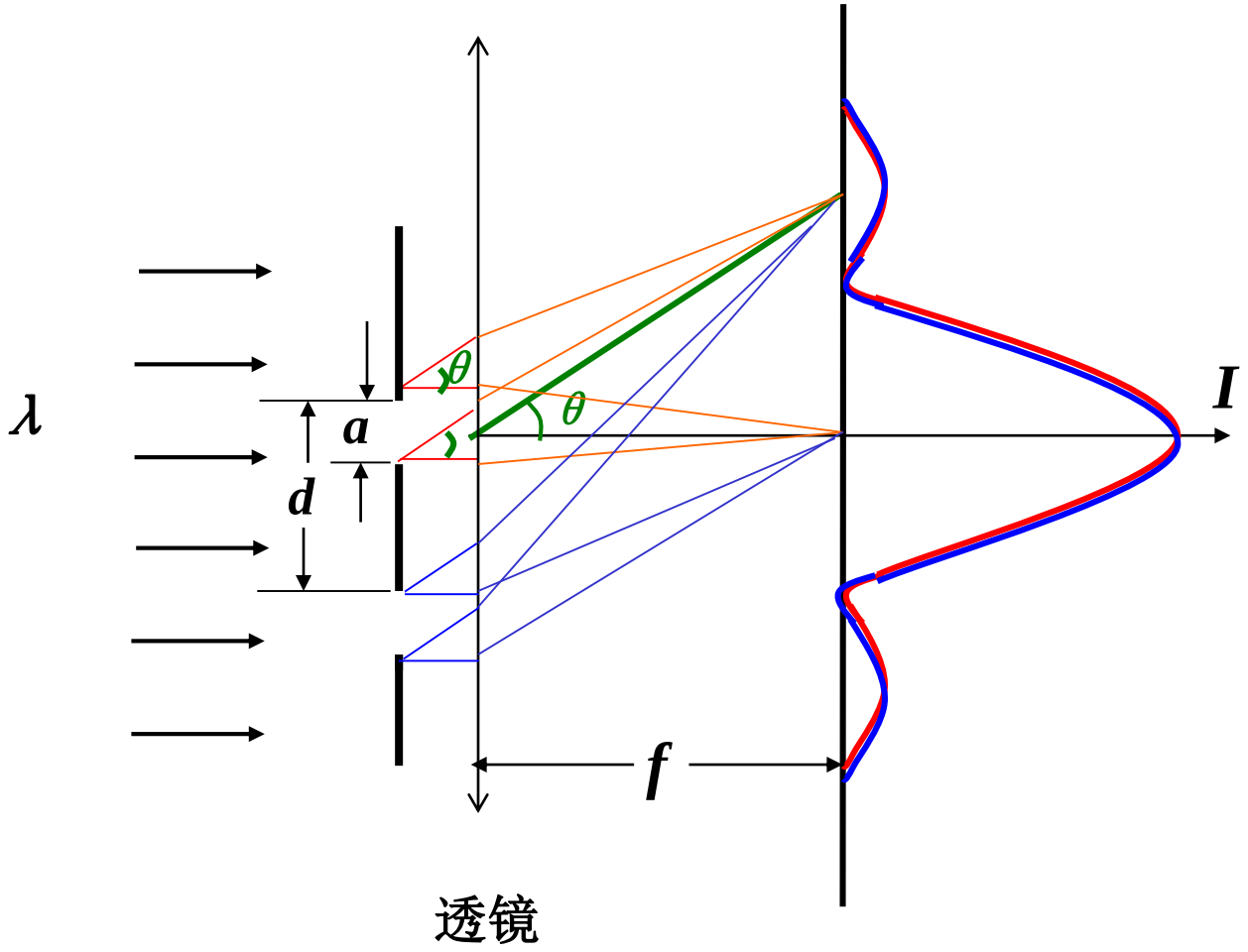
一．衍射对双缝干涉的影响

不考虑衍射时，双缝干涉的光强分布图：

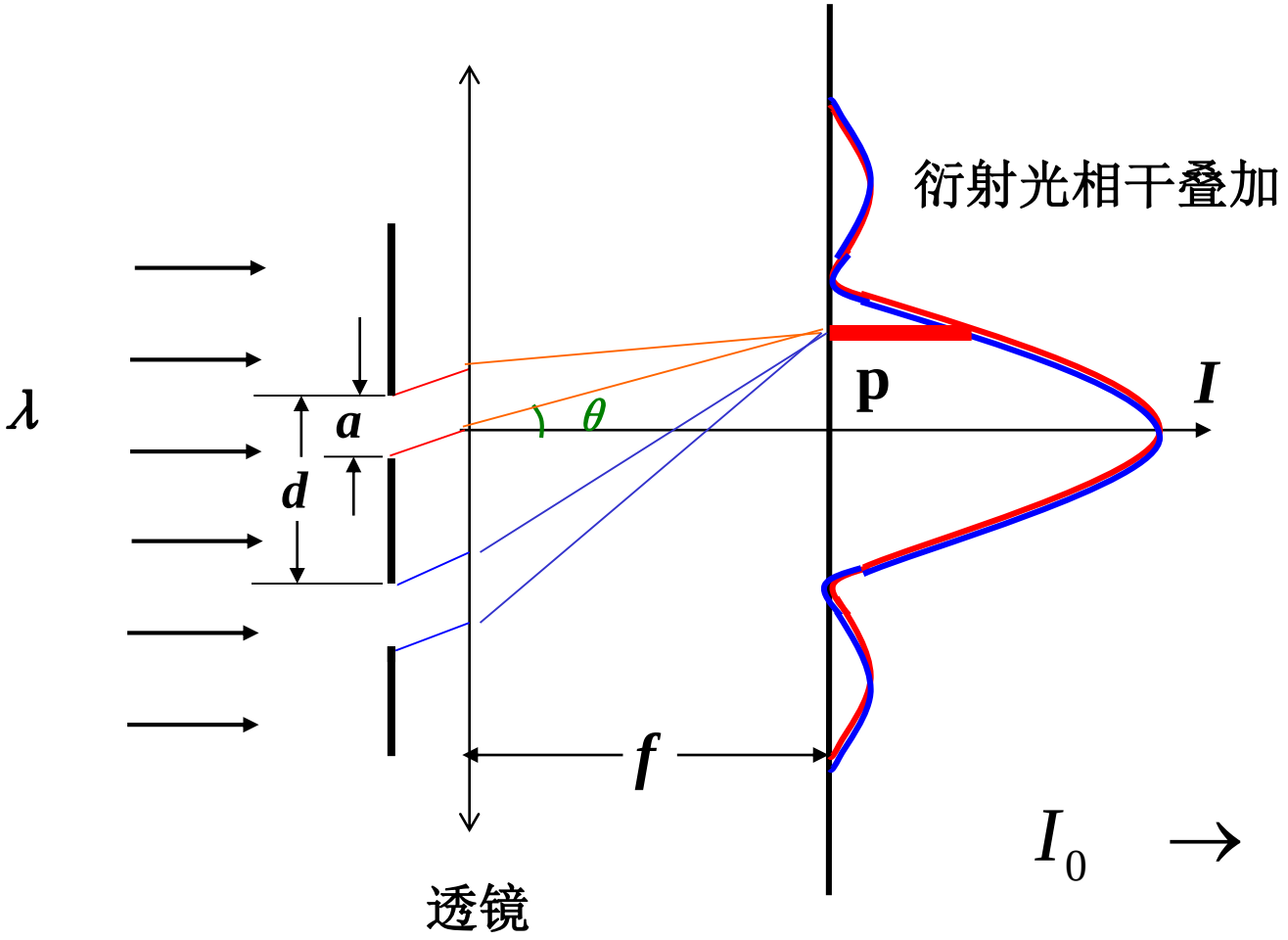


明纹　$d\sin\theta = \pm k\lambda$

假设了双缝中每个缝单独存在时，光强均匀分布



设双缝的每个缝宽均为 a，在夫琅禾费衍射下，每个缝的衍射图样位置是相重叠的。



$$I_0 \rightarrow I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$$

$$I = 2I_0(1+\cos\Delta\varphi)$$

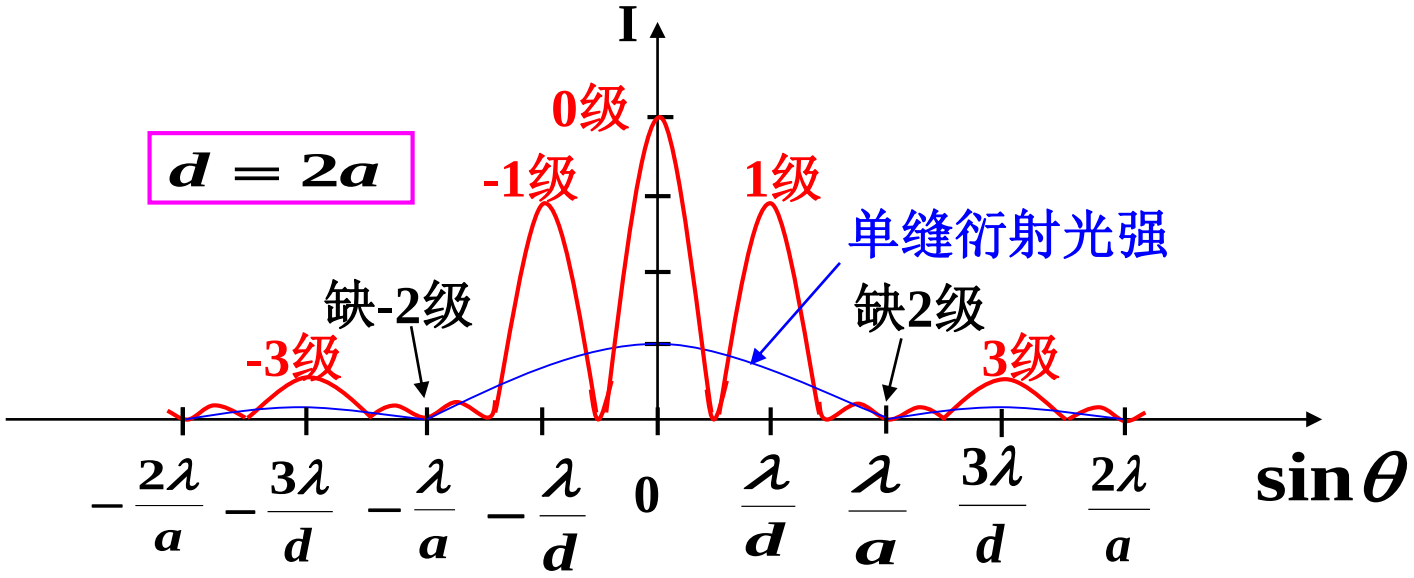
$$\Delta\varphi = \frac{2\pi d\sin\theta}{\lambda}$$

$$\boxed{I = 2I_0\left(\frac{\sin\alpha}{\alpha}\right)^2(1+\cos\Delta\varphi)}$$

双缝干涉条纹各级主极大的强度不再相等，而是受到了衍射的调制.

$$I_0 \quad \rightarrow \quad I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$$

$d = 2a$ 时，双缝干涉光强受衍射调制如下图



- 明纹缺级现象

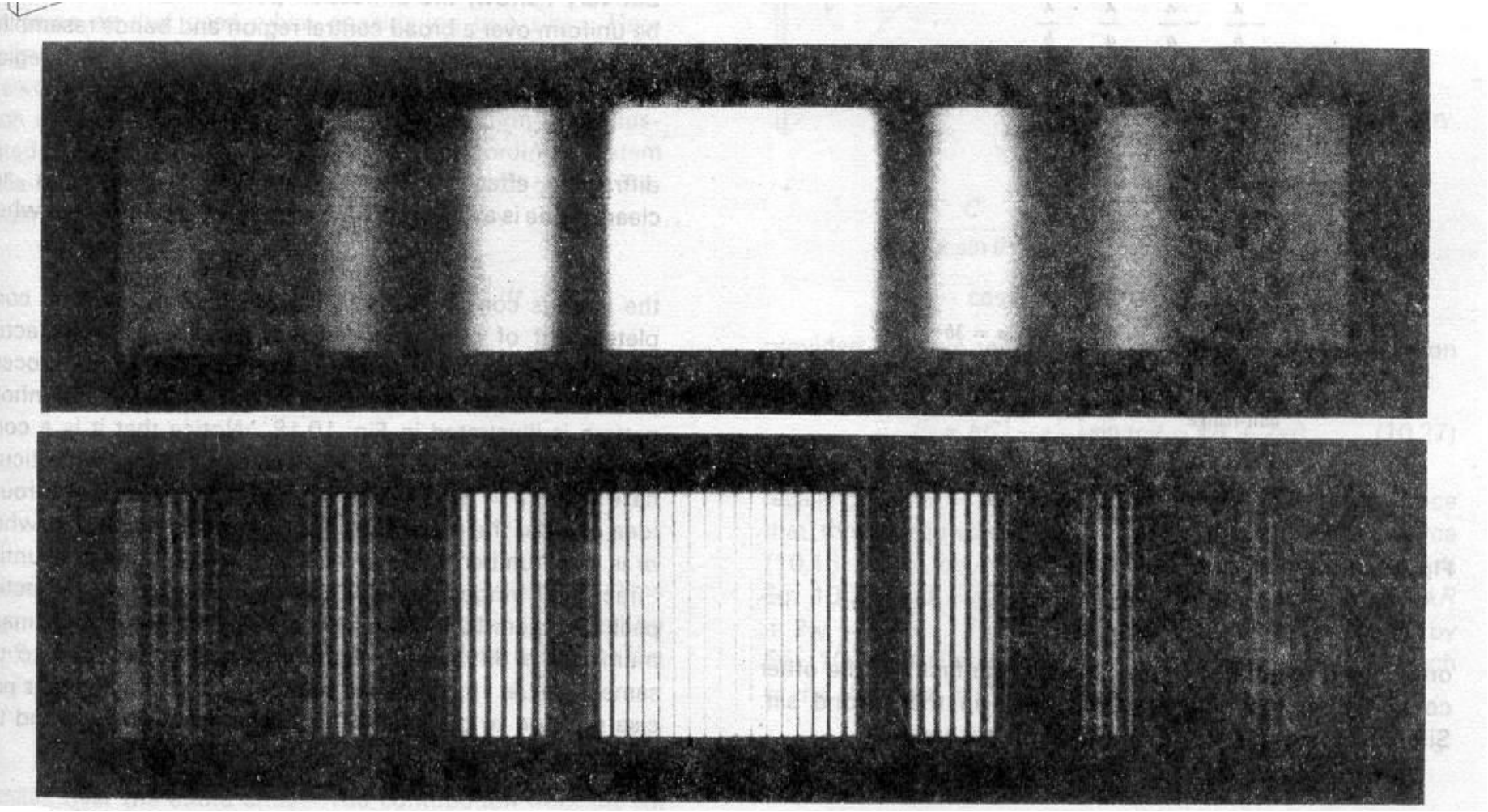
干涉明纹位置： $d\sin\theta = \pm k\lambda$， $k = 0, 1, 2, \cdots$

衍射暗纹位置： $a\sin\theta' = \pm k'\lambda$， $k' = 1, 2, 3, \cdots$

$\dfrac{d}{a} = \dfrac{k}{k'}$ 时 $\theta = \theta'$， 出现缺级。

干涉明纹缺级级次 $k = \dfrac{d}{a}k'$

- $\dfrac{d}{a}$ 较大时的现象

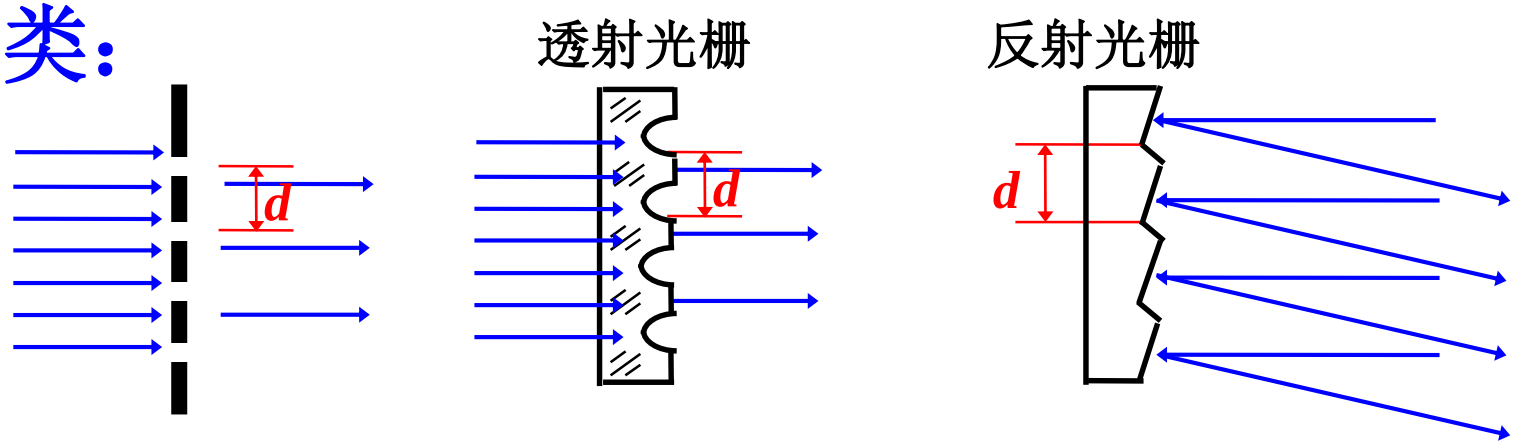


单缝(上)和双缝(下)衍射图案

二. 光栅

1. 光栅—大量等宽等间距的平行狭缝(或反射面)构成的光学元件。

2. 种类:



3. 光栅常数

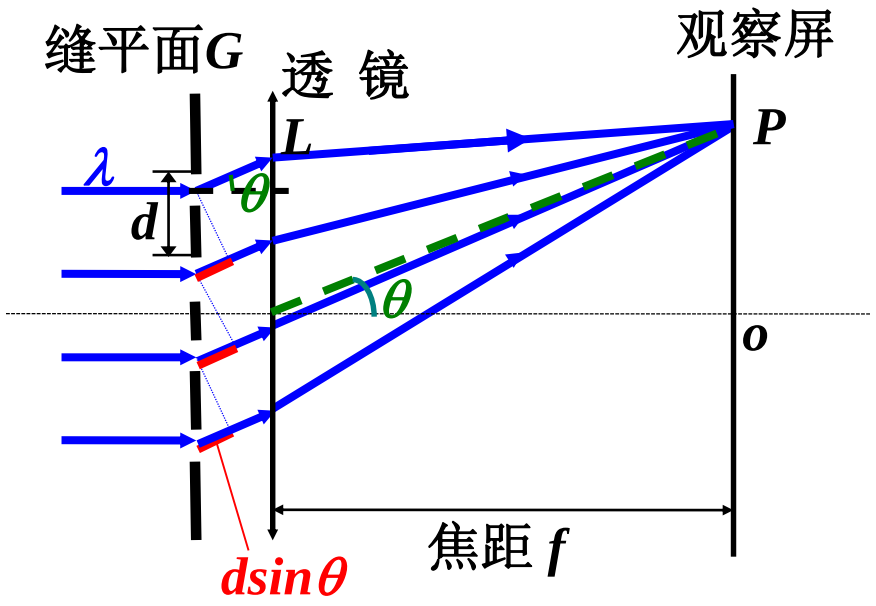
a 是透光（或反光）部分的宽度

b 是不透光(或不反光)部分的宽度

$d = a+b$ 光栅常数（缝间距）

三. 光栅衍射

1. 多光束干涉



明纹(主极大)条件：

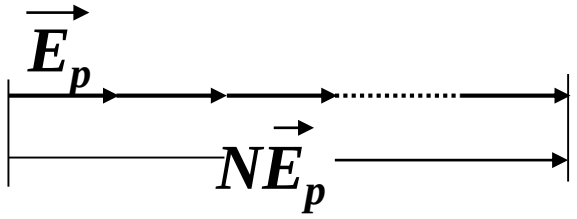
$$d\sin\theta = \pm k\lambda$$

$$k = 0, 1, 2, 3\cdots$$

光栅方程

设每个缝发的光在对应衍射角θ 方向的P点的光振动的振幅为E_p

P点为主极大时 $\Delta\varphi = \pm 2k\pi$



$$I_P \propto N^2 E_p^2$$

主极大间距 $\Delta\theta = \dfrac{\lambda}{d\cos\theta}$

暗纹条件： N个矢量闭合

$$N\Delta\varphi = \pm 2k\pi \qquad (1) \quad k = 1, 2, \cdots \neq Nm$$

又

$$\Delta\varphi = \frac{2\pi}{\lambda}\cdot d\cdot\sin\theta \qquad (2)$$

由(1)，(2)得

$$d\sin\theta = \frac{\pm k}{N}\lambda \qquad (k \neq Nm \neq 0)$$

$m, m+1$亮纹之间

$$k = Nm+1, Nm+2, \cdots, Nm+N-1$$

暗纹间距= $\dfrac{\text{主极大间距}}{N}$ $\longrightarrow$ $\delta\theta = \dfrac{\lambda}{Nd\cos\theta}$

相邻主极大间有$N-1$个暗纹

相邻主极大间有$N-2$个次极大

如 $N = 4$, 0和1级亮纹之间有三个极小

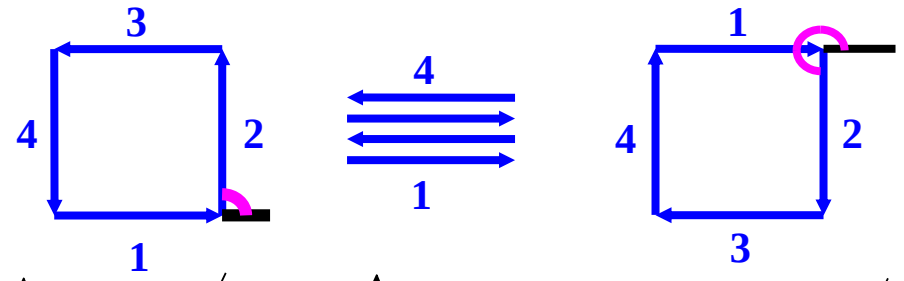
$$\sin\theta = \frac{1}{4}\cdot\frac{\lambda}{d} \quad , \quad \frac{2}{4}\cdot\frac{\lambda}{d} \quad , \quad \frac{3}{4}\cdot\frac{\lambda}{d}$$

$$(k=1)\ ,\ (k=2)\ ,\ (k=3)$$

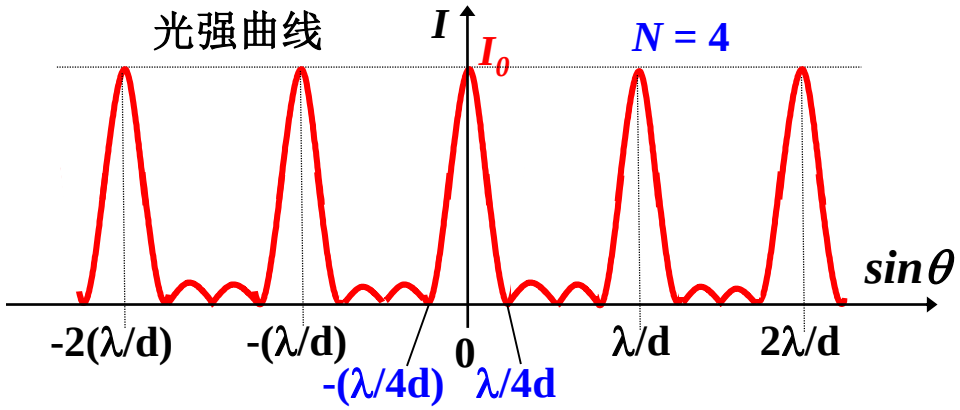
$$d\sin\theta = \frac{\pm k}{N}\lambda$$

$$\Delta\varphi = \frac{d\cdot\sin\theta}{\lambda}\cdot 2\pi$$

$\Delta\varphi =$ $\pi/2$, π, $3\pi/2$

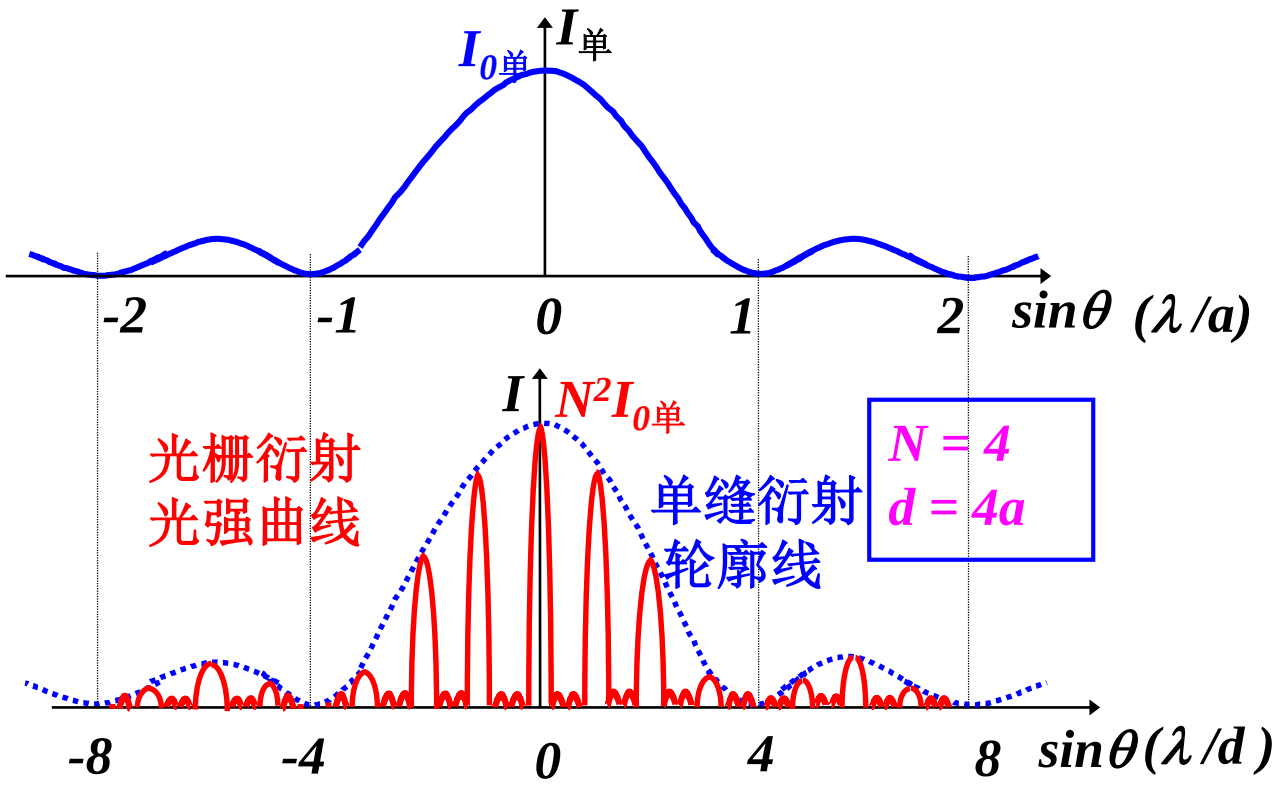


$\Delta\varphi = \pi/2$ $\Delta\varphi = \pi$ $\Delta\varphi = 3\pi/2$



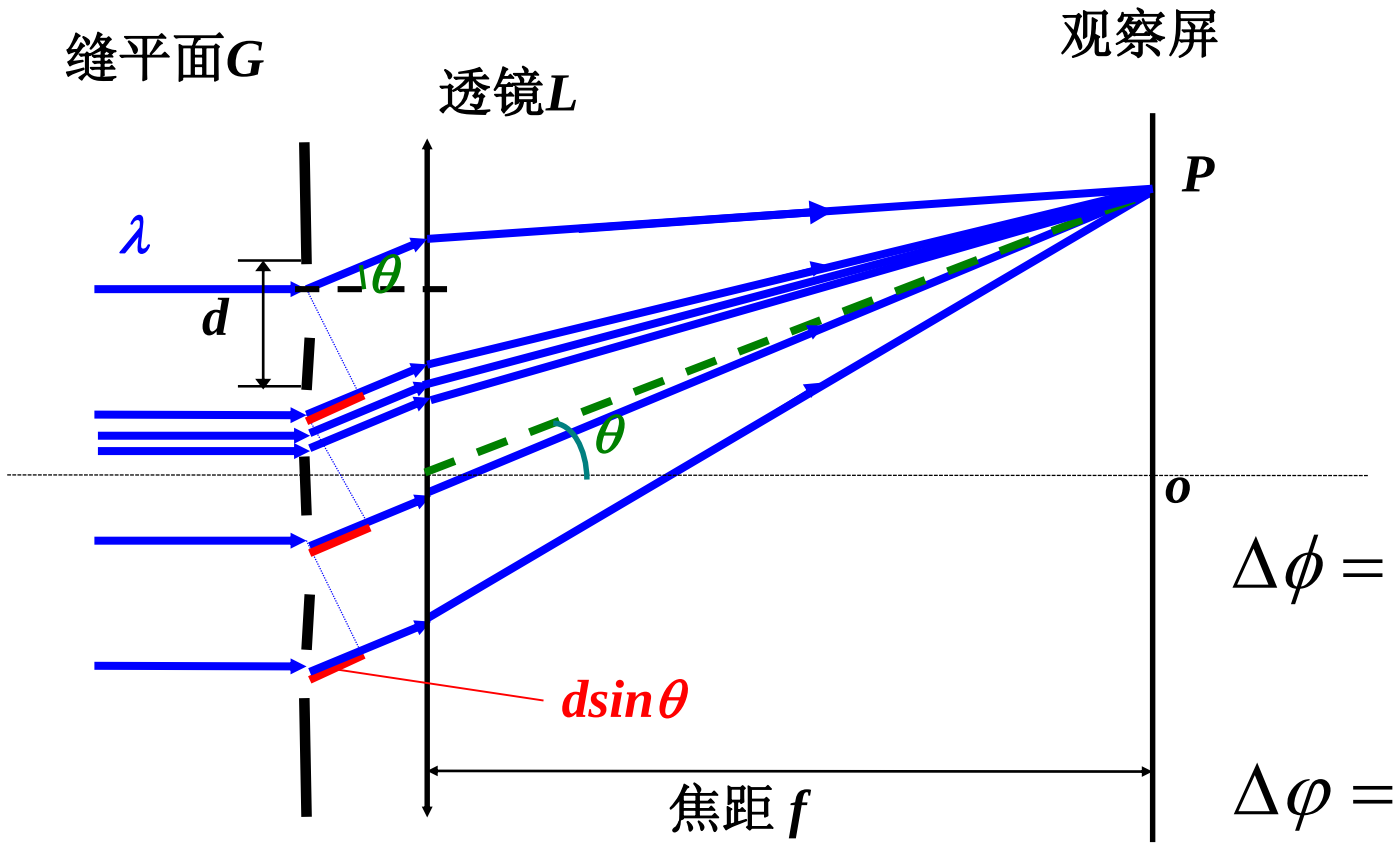
2. 光栅衍射

（1）各主极大受到单缝衍射的调制 $\left(\frac{\sin\alpha}{\alpha}\right)^2$



（2）$\frac{b}{a}$ 为整数比时，会出现缺级。

3. 光栅夫琅禾费衍射的光强公式



$$\Delta\phi = \frac{2\pi}{\lambda}\frac{a\sin\theta}{M}$$

$$\Delta\varphi = \frac{2\pi}{\lambda}d\sin\theta$$

$$A_P = \Delta E_0 \sum_{n=1}^{N}\sum_{m=1}^{M}\cos[\omega t + (m-1)\Delta\phi + (n-1)\Delta\varphi]$$

$$A_P = E_{\mathrm{p}} \sum_{n=1}^{N} \cos[\omega t + \alpha + (n-1)\Delta\varphi]$$

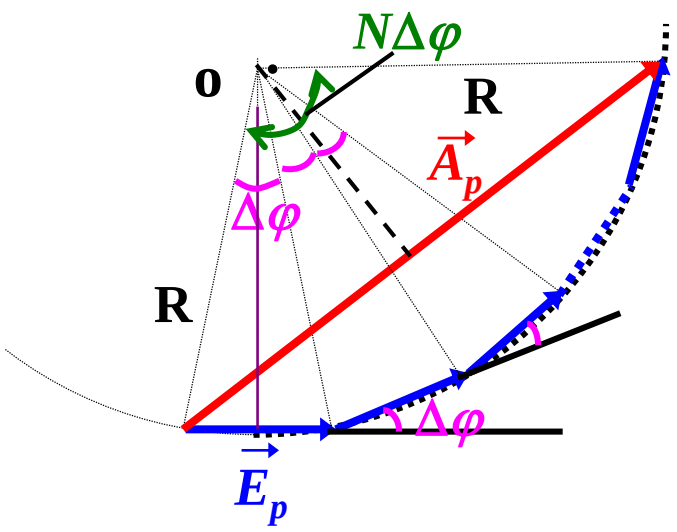
$$E_{\mathrm{p}} = E_{0单} \frac{\sin\alpha}{\alpha} \quad , \qquad \alpha = \frac{\pi}{\lambda} a \sin\theta$$

相邻缝在P点

$$\Delta\varphi = \frac{2\pi}{\lambda} d \sin\theta$$

$$A_p = 2R \sin\frac{N\Delta\varphi}{2}$$

$$E_{\mathrm{p}} = 2R \sin\frac{\Delta\varphi}{2}$$



$$A_p = E_p \sin(N\frac{\Delta\varphi}{2}) / \sin\frac{\Delta\varphi}{2}$$

$$\therefore\ A_p = E_p \cdot \frac{\sin N\dfrac{\Delta\varphi}{2}}{\sin\dfrac{\Delta\varphi}{2}} = E_{0单} \cdot \frac{\sin\alpha}{\alpha} \cdot \frac{\sin N\beta}{\sin\beta}$$

$$\beta = \frac{\Delta\varphi}{2} = \frac{\pi d}{\lambda} \cdot \sin\theta$$

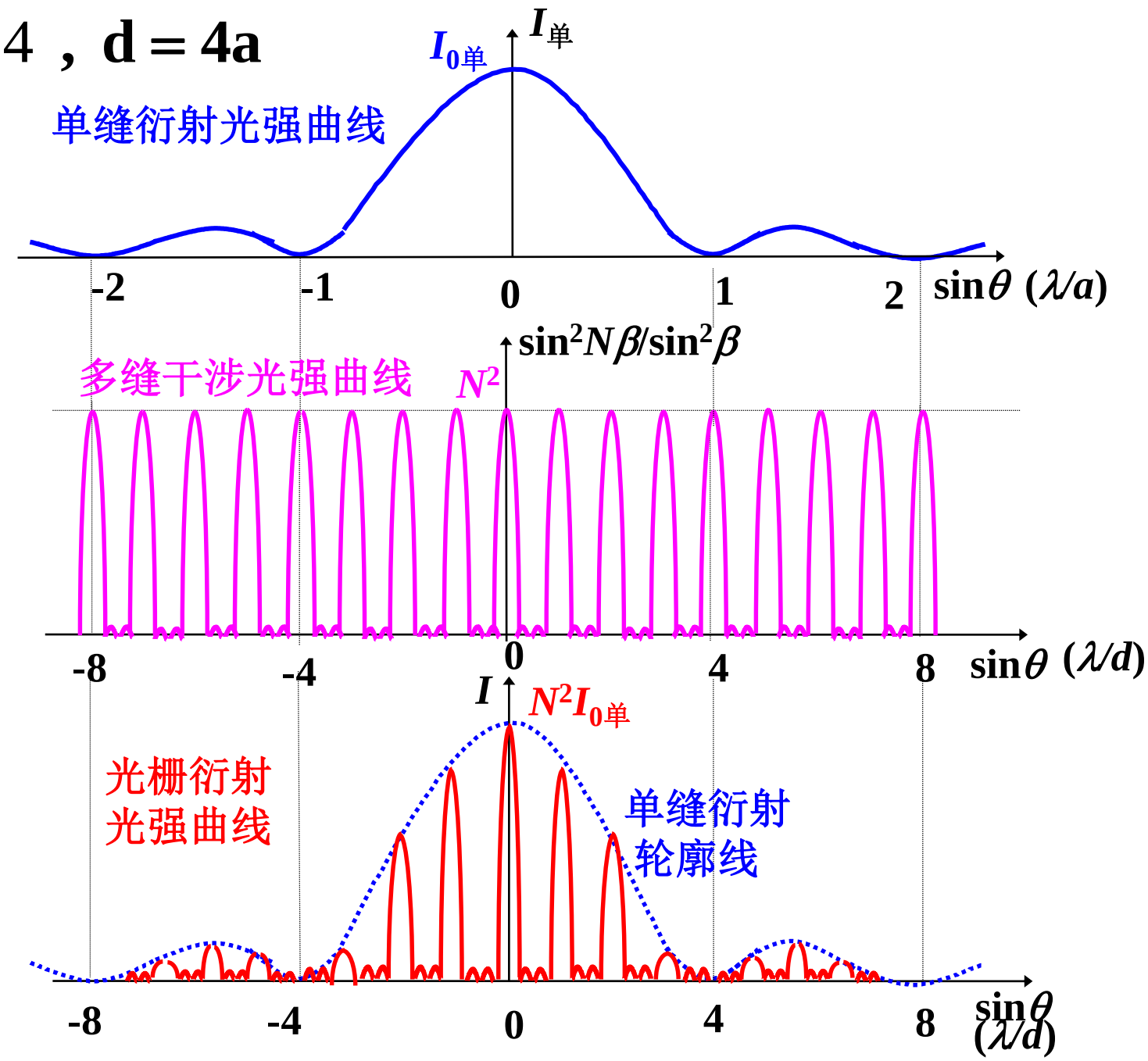
单缝中央主极大光强

$$I_p = I_{0单}\left(\frac{\sin\alpha}{\alpha}\right)^2 \cdot \left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

单缝衍射因子

多缝干涉因子

$N = 4$ ， $d = 4a$


$I_{单}$
$I_{0单}$
单缝衍射光强曲线
-2
-1
0
1
2
$\sin\theta$ (λ/a)
$\sin^2 N\beta/\sin^2\beta$
多缝干涉光强曲线
N^2
-8
-4
0
4
8
$\sin\theta$ (λ/d)
I
$N^2 I_{0单}$
光栅衍射
光强曲线
单缝衍射
轮廓线
-8
-4
0
4
8
$\sin\theta$
(λ/d)

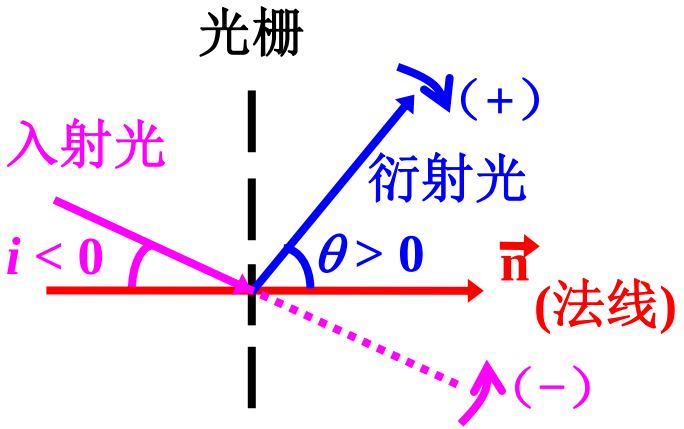
四. 斜入射的光栅方程、相控阵雷达

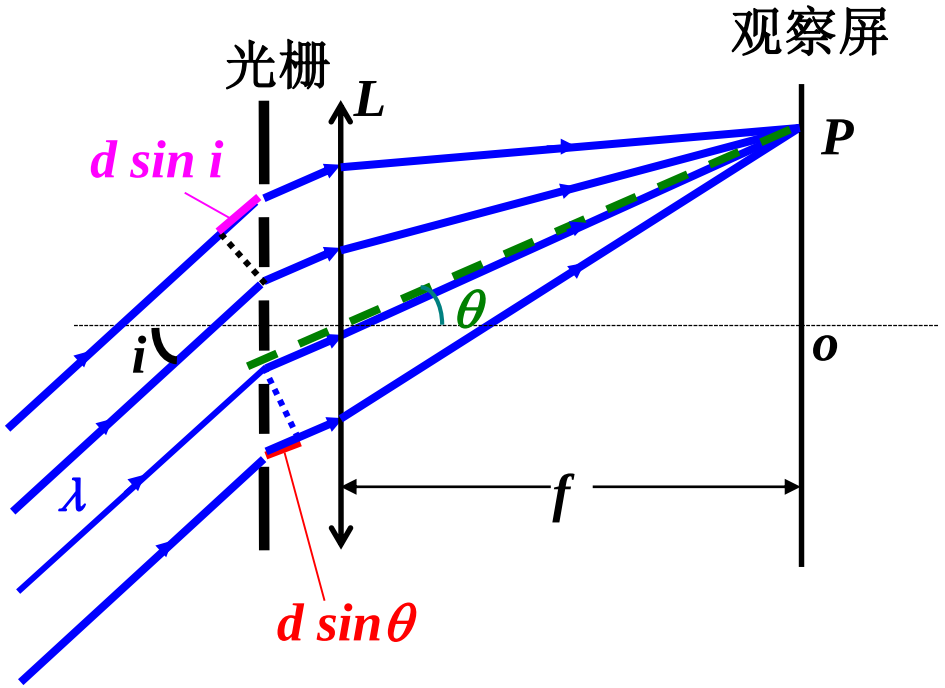
1. 光线斜入射时的光栅方程

$$\delta = d(\sin\theta - \sin i)$$

$$d(\sin\theta - \sin i) = \pm k\lambda$$

i 和 ***θ*** 的符号规定:





k 确定时，调节***i***，则***θ*** 相应改变.

例如，令 $k=0$，则 $d\cdot\sin\theta = d\cdot\sin i$

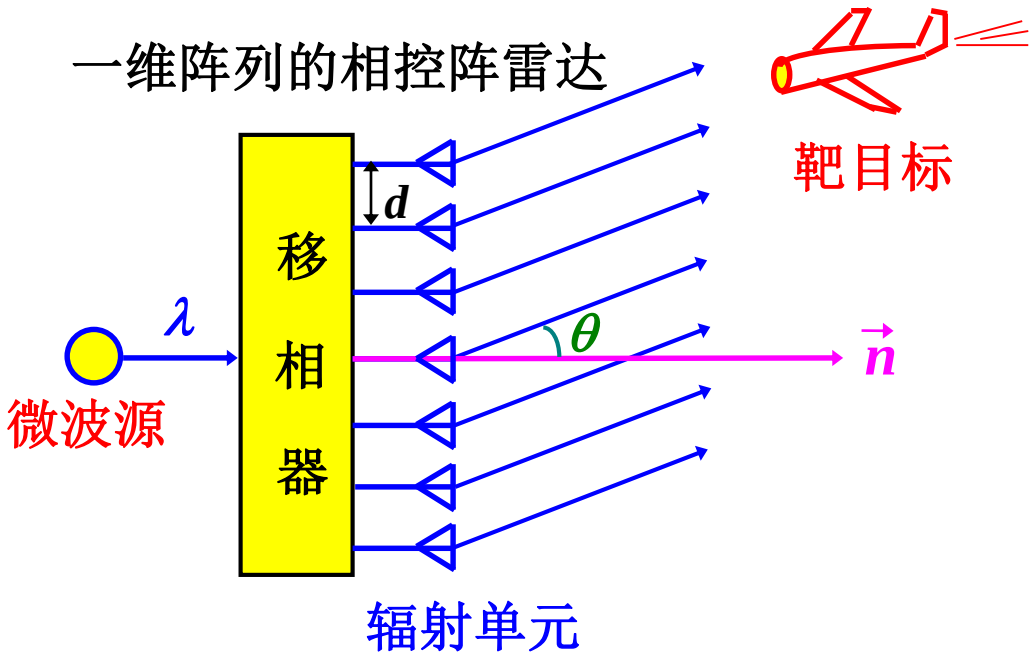
$$\Delta\varphi = \frac{d\cdot\sin i}{\lambda}\cdot 2\pi = \frac{d\cdot\sin\theta}{\lambda}\cdot 2\pi$$

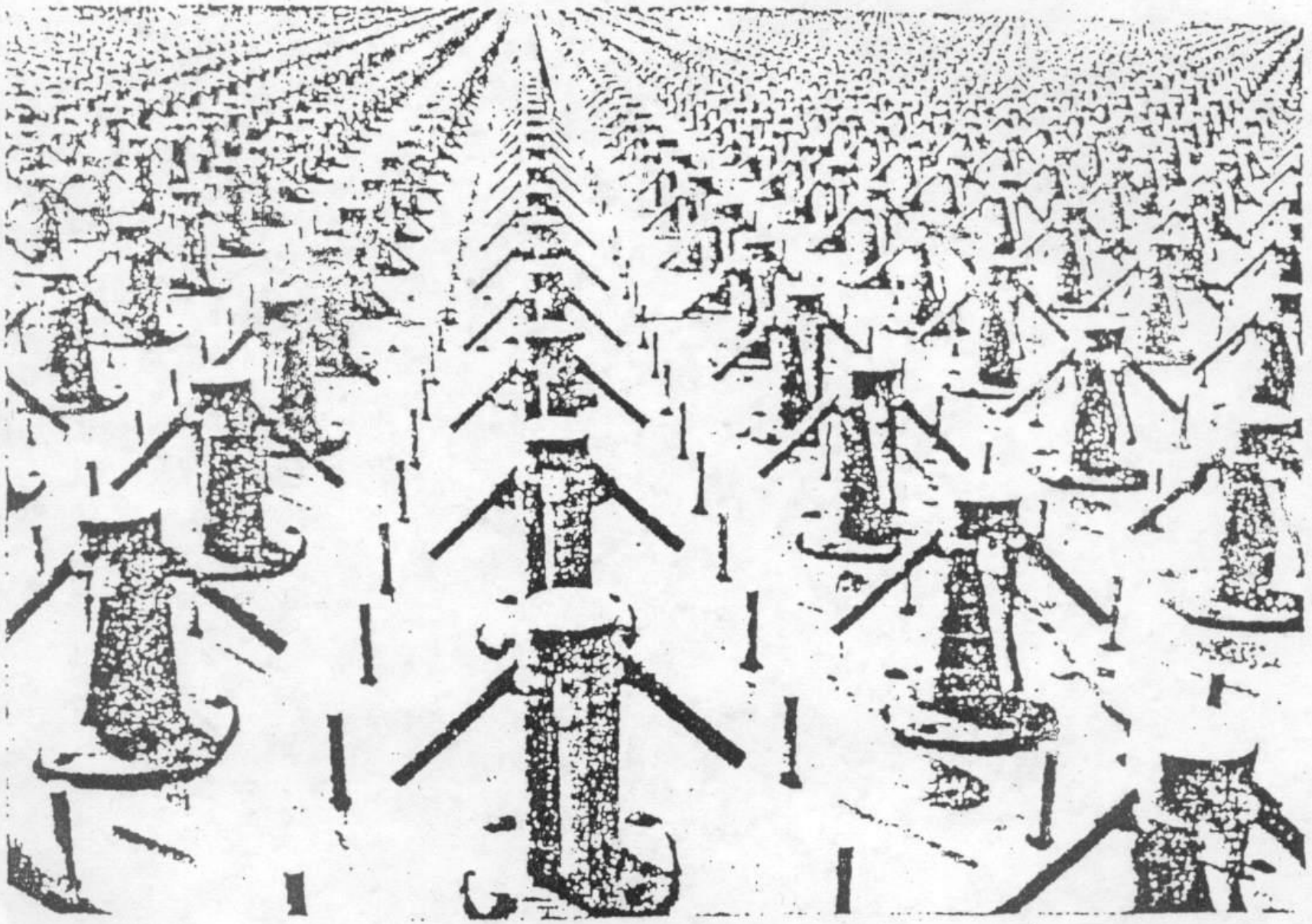
$$\therefore\quad \sin\theta = \frac{\lambda}{2\pi d}\cdot\Delta\varphi$$

改变相邻入射光的相位差
即可改变0级衍射光的方向

2. 相控阵雷达

- 相位控制扫描
- 频率控制扫描





单选题 1分

设置

用每毫米N=400条刻痕的衍射光栅观察λ=500 nm (1 nm=10^{-9} m)的光谱线，入射角θ=30°能观测到谱线的最高级次k为：

A 9

B 8

C 7

D 6

E 5

提交

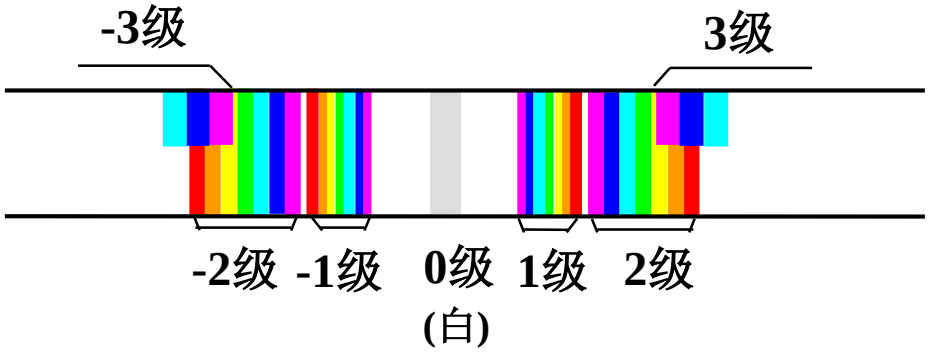
§4　光学仪器的分辨本领

一．光栅光谱，光栅的色散本领、分辨本领

1．光栅光谱

$$d\sin\theta = \pm k\lambda$$

白光的光栅光谱



光栅光谱有多级，且是正比光谱

根据亮纹位置可以确定光的波长及其光强

H

Na

Ne

Hg

H_2

太阳光谱

2. 光栅的色散本领

把不同波长的光在谱线上分开的能力

设： 波长 λ 的谱线，亮纹 θ ，位置 x；
波长 $\lambda+\delta\lambda$ 的谱线，亮纹 $\theta+\delta\theta$ ， 位置 $x+\delta x$

角色散本领 $D_\theta \equiv \dfrac{\delta\theta}{\delta\lambda}$

线色散本领 $D_l \equiv \dfrac{\delta x}{\delta\lambda}$

$$\boxed{D_l = f \cdot D_\theta}$$

f—光栅后的透镜焦距

k 级 亮纹 $\sin\theta - \sin i = k\dfrac{\lambda}{d}$

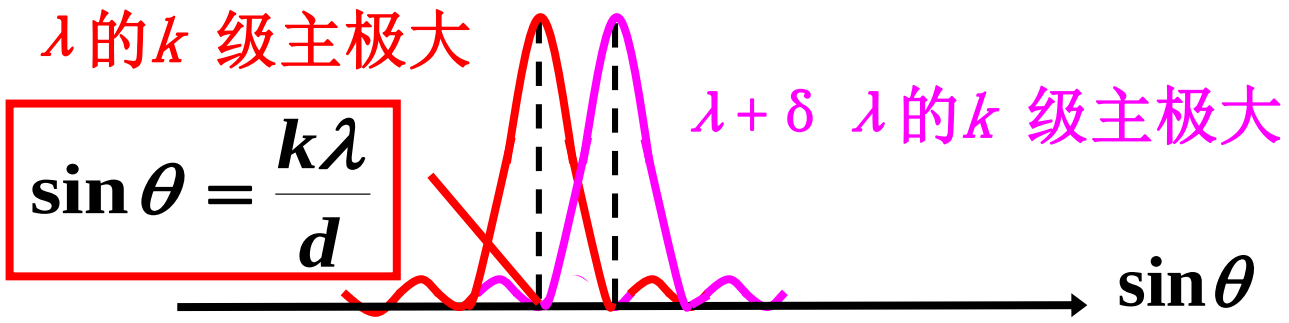
由 $\cos\theta \cdot \delta\theta = k\dfrac{\delta\lambda}{d}$

有 $\boxed{D_\theta = \dfrac{k}{d \cdot \cos\theta}}$ $\boxed{D_l = \dfrac{k \cdot f}{d \cdot \cos\theta}}$

3. 光栅的色分辨本领

设入射波长为λ和λ+δλ时，二者的谱线刚能分开

光栅分辨本领 $\boxed{R \equiv \dfrac{\lambda}{\delta\lambda}}$



$$\boxed{\sin\theta = \frac{k\lambda}{d}}$$

瑞利判据:相邻的两个等光强的非相干亮纹，如果其一个亮纹的中心恰好落在另一亮纹的边缘暗纹, 则此两亮纹被认为是刚刚可以分辨。

对应 $k' = Nk - 1$ 的 $(\lambda + \delta\lambda)$ 的暗纹，$\sin\theta = \dfrac{k'(\lambda + \delta\lambda)}{Nd}$

$$\frac{k}{d} \cdot \lambda = \frac{Nk - 1}{Nd} \cdot (\lambda + \delta\lambda)$$

$$\frac{k}{d}\delta\lambda - \frac{\lambda+\delta\lambda}{Nd} = 0 \qquad kN - 1 = \frac{\lambda}{\delta\lambda}$$

得

$$\boldsymbol{R = \frac{\lambda}{\delta\lambda} = Nk - 1 \approx Nk, (k \neq 0)} \quad \therefore \left.\begin{matrix}\uparrow N \\ \uparrow k\end{matrix}\right\} \rightarrow \uparrow \boldsymbol{R}$$

$$(N >> 1)$$

例3. 对Na双线： λ_1 =589nm ,

λ_2 =λ+δλ=589.6nm

$$R = \frac{\lambda}{\delta\lambda} = \frac{589}{0.6} \approx 982 = Nk$$

k=1, N > 982 可分辨开Na双线

单选题 1分

设置

用每毫米N=550条刻痕的衍射光栅来观察正入射的钠黄光（λ=589 nm和589.6 nm ）的第2级光谱线，双黄光分离的角度为

A 4.0×10^{-4} rad

B 4.6×10^{-4} rad

C 6.4×10^{-4} rad

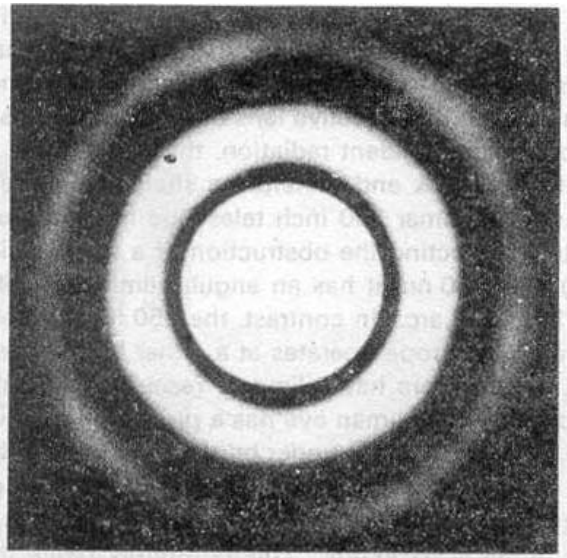
D 7.4×10^{-4} rad

E 以上答案都不对

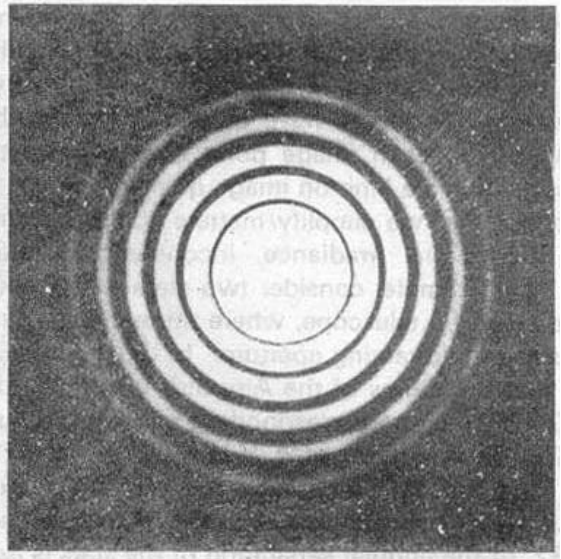
提交

二. 透镜的分辨本领

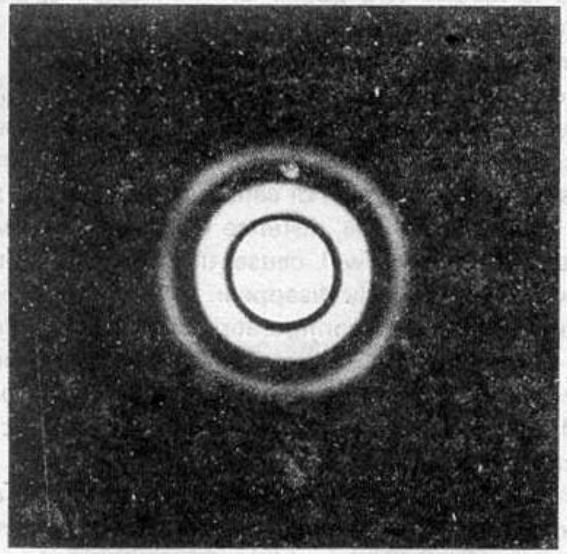
圆孔的夫琅禾费衍射



Airy rings (0.5 mm hole diameter). [Photo by E.H.]



(a)

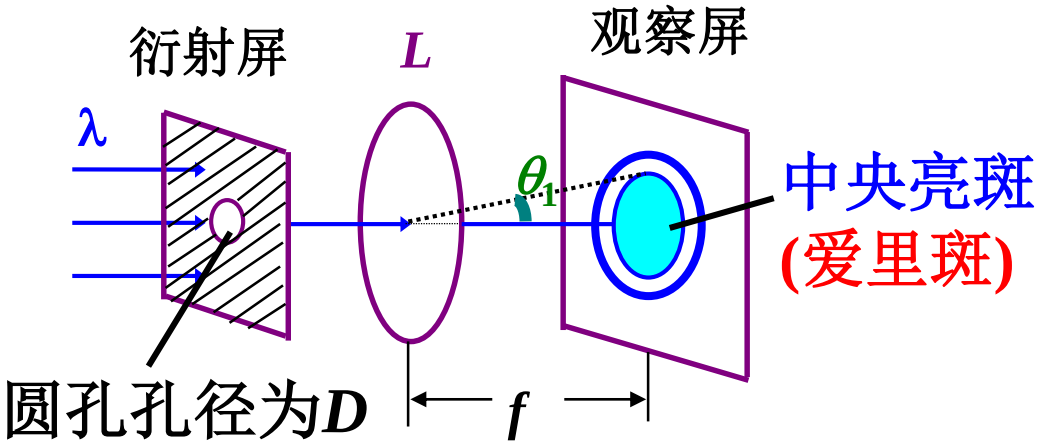


Airy rings (1.0 mm hole diameter). [Photo by E.H.]

(b)

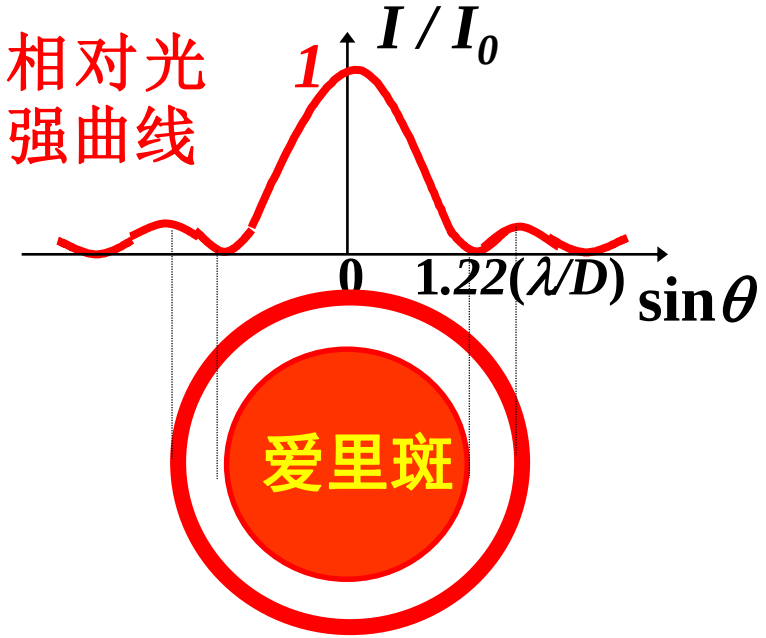
(a) Airy rings—long exposure (1.5 mm hole diameter). (b) Central Airy disc—short exposure with the same aperture. [Photos by E.H.]

1. 圆孔的夫琅禾费衍射

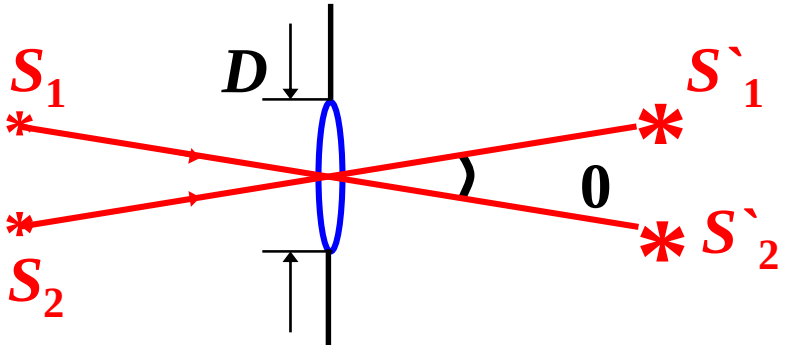


$$D \cdot \sin\theta_1 \approx 1.22\lambda$$

$$\left.\begin{matrix} D\uparrow \\ \lambda\downarrow \end{matrix}\right\} \text{爱里斑变小}$$



2. 透镜的分辩本领



几何光学 ：（ 经透镜成像 ）

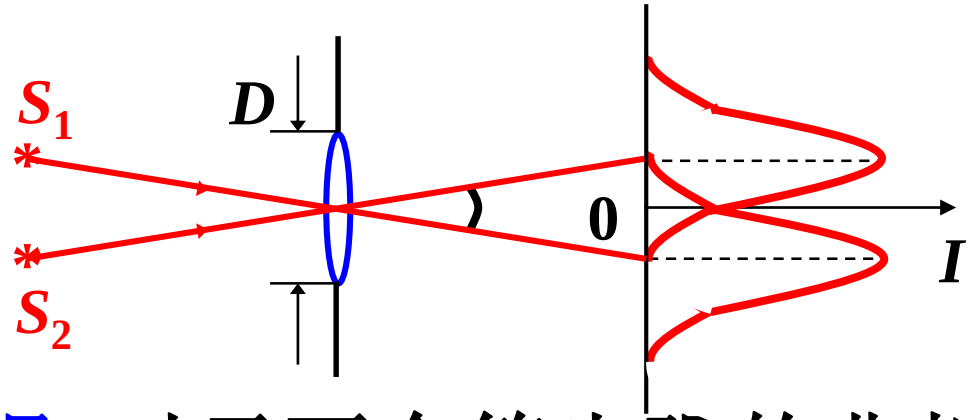
物点 ⇒ 象点

物(物点集合) ⇒ 象(象点集合)

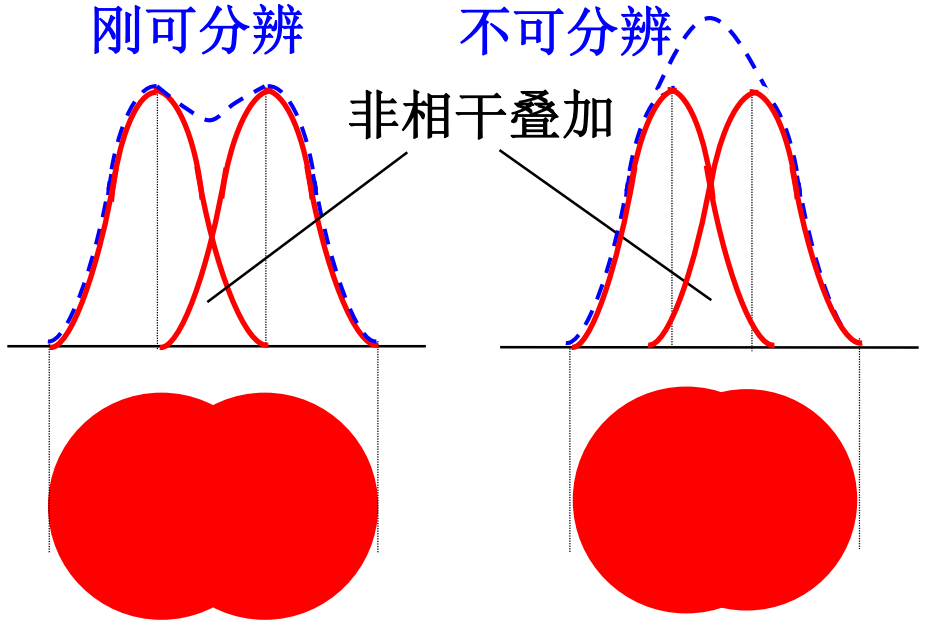
波动光学 ：（ 经透镜成像 ）

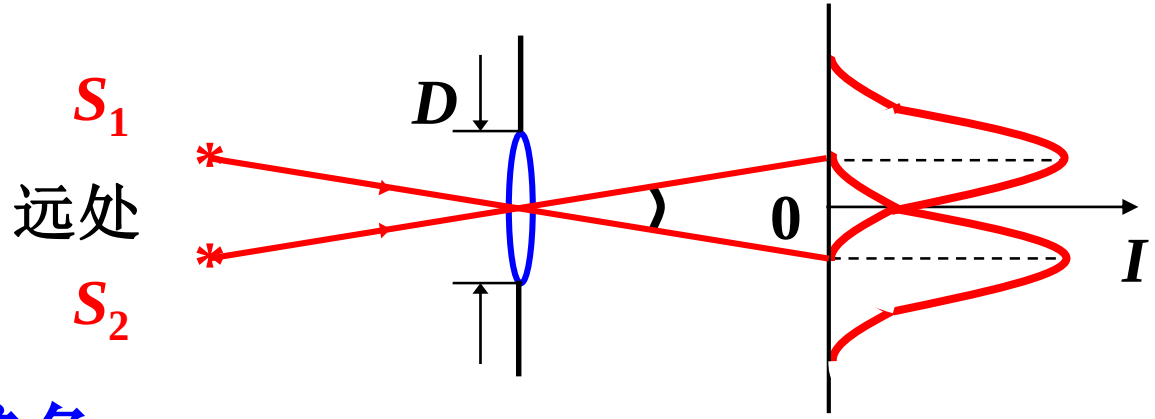
物点 ⇒ 象斑(爱里斑)

物(物点集合) ⇒ 象(象斑集合)



瑞利判据：对于两个等光强的非相干物点，如果其一个象斑的中心恰好落在另一象斑的边缘（第一暗纹处），则此两物点被认为是刚刚可以分辨。





最小分辨角

$$\delta\theta = \theta_1 \approx 1.22\frac{\lambda}{D}$$

分辨本领

$$R \equiv \frac{1}{\delta\theta} = \frac{D}{1.22\lambda} \qquad \left.\begin{matrix} D\uparrow \\ \lambda\downarrow \end{matrix}\right\} \rightarrow R\uparrow$$

人眼分辨本领

瞳孔直径2-4毫米 $\delta\theta = \theta_1 \approx 1.22\frac{\lambda}{D} \sim 0.2$毫弧度

明视距离25厘米　　最小分辨角 ~0.2毫弧度

分辨最小距离 ~ 0.05毫米

长城宽~ 10米　　太空高度~ 300公里

角宽度 ~ 0.03毫弧度

地月距离 38万公里多

能分辨的物体大小 ~ 76公里

望远镜：λ不可选择，可 $\uparrow D \rightarrow \uparrow R$

显微镜：D不会很大，可 $\downarrow \lambda \rightarrow \uparrow R$

单选题 1分

设汽车前大灯的距离为d=1.9m，波长为λ=550μm。人眼瞳孔直径为D=5mm，根据瑞利判据，人眼刚刚能分辨两只大灯时，汽车的距离L为

A 11km

B 13km

C 14km

D 15km

E 以上答案都不对

设置

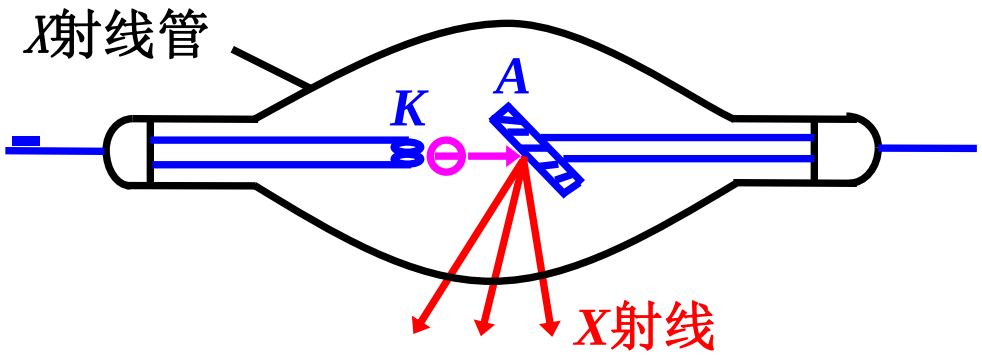
提交

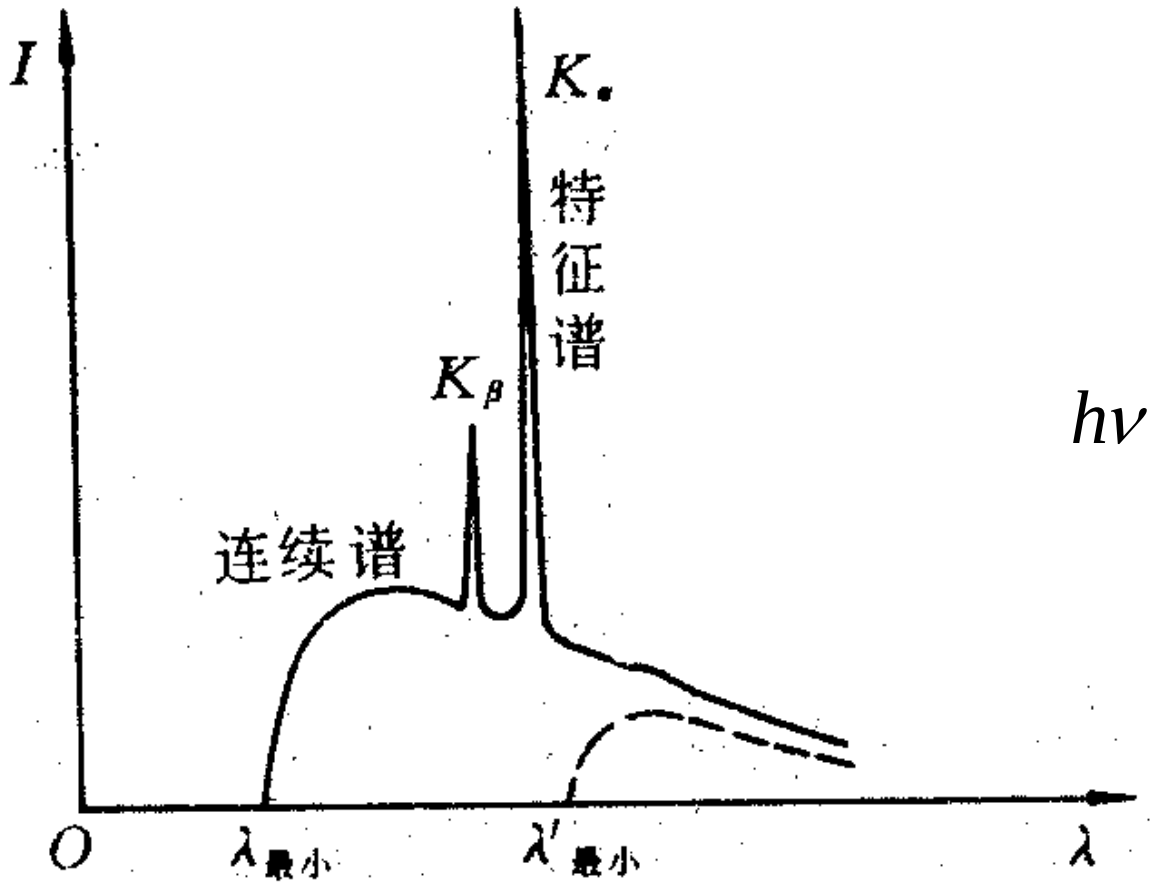
§5　X射线的衍射

X射线的发现：机遇与探索精神

一．X射线的产生

X射线 λ ： 10^{-2}—10^{1}nm



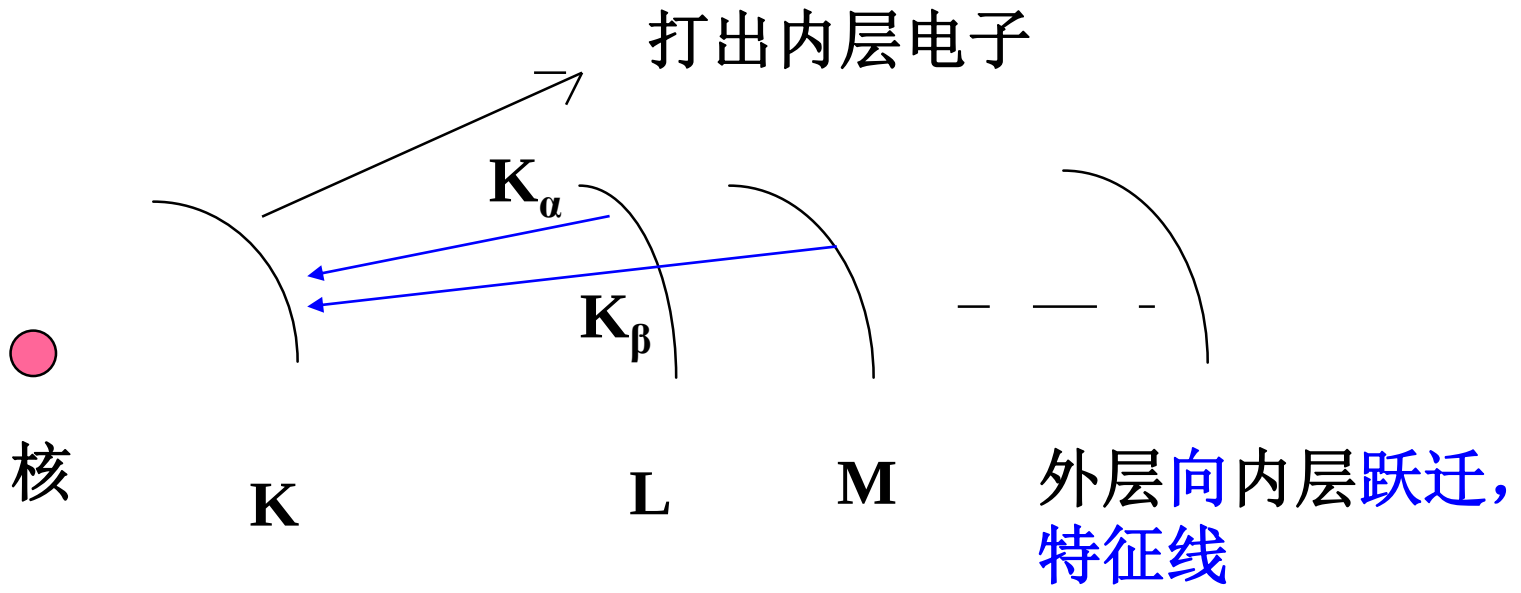


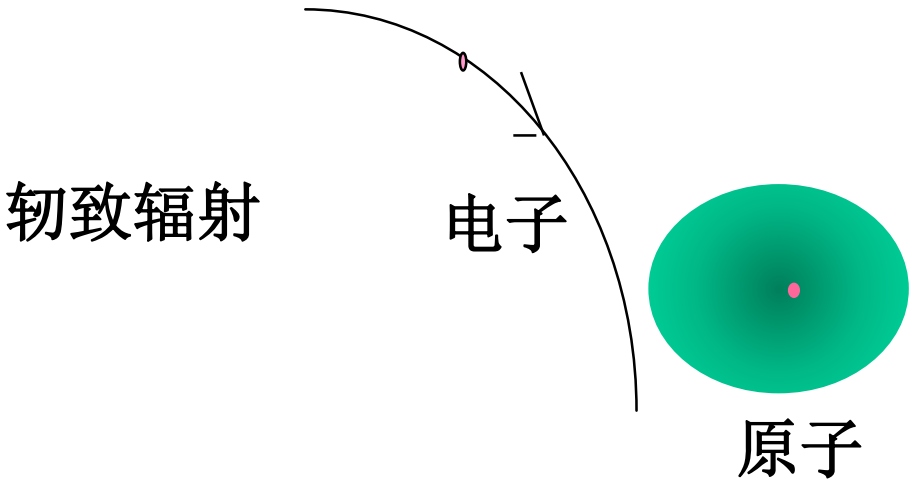
$$h\nu = \frac{hc}{\lambda}$$

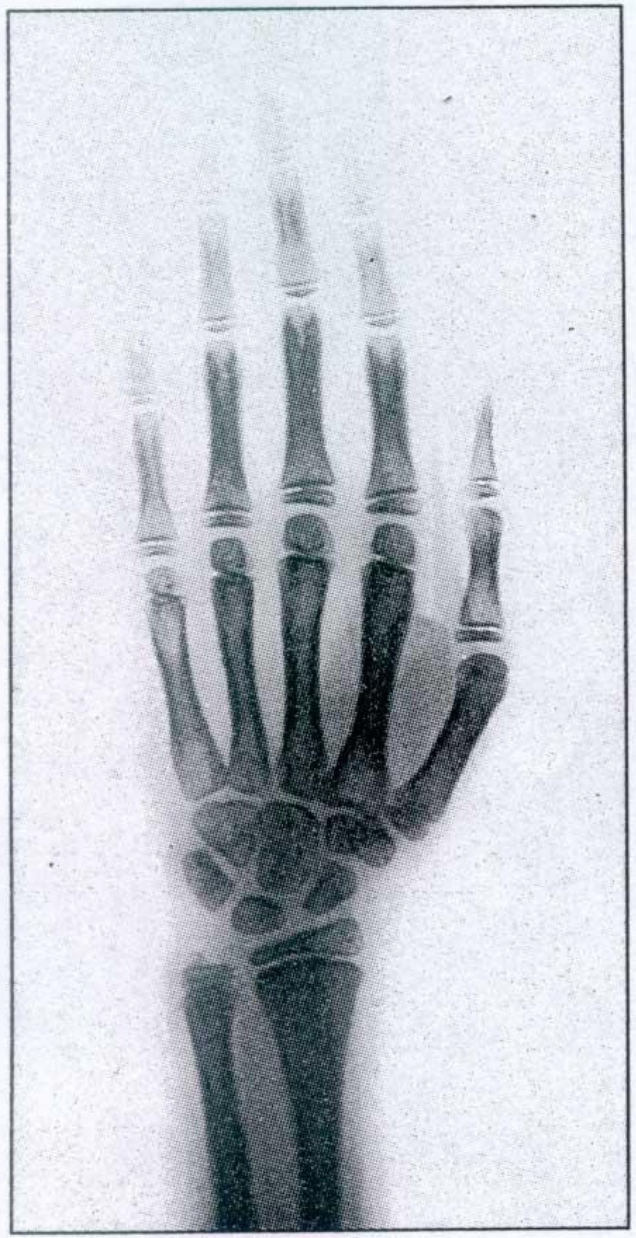
X 射线谱

波长最小值对应电子最高能量

- 特征谱和连续谱



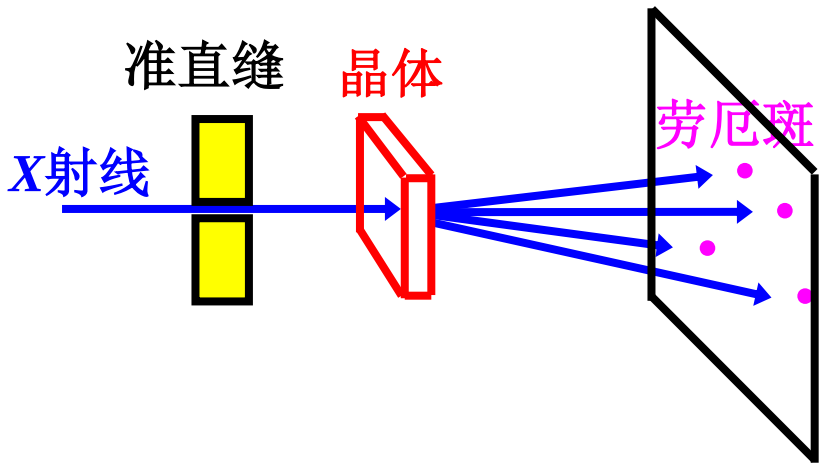




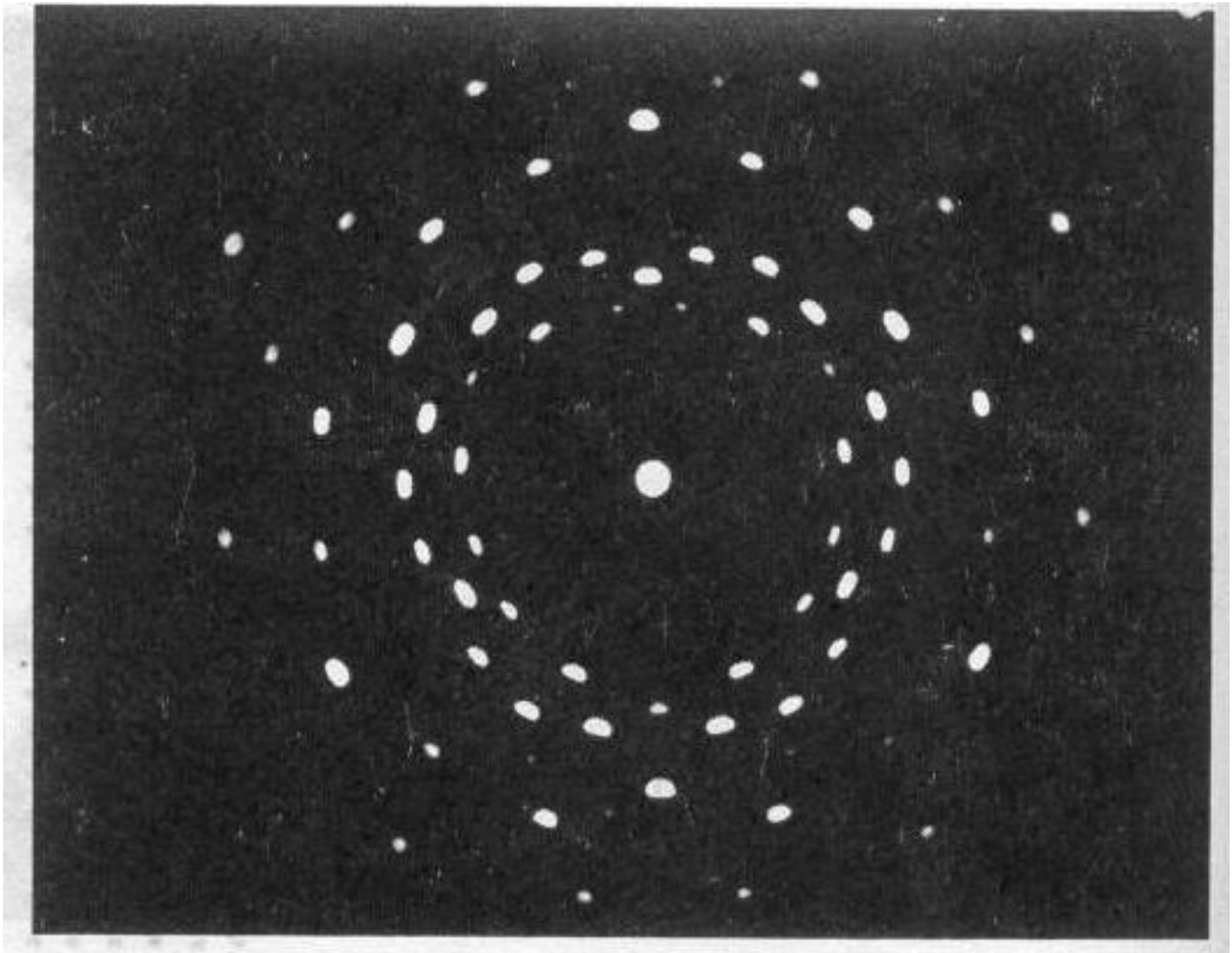
X-ray photograph of a human hand.

劳厄（Laue）实验（1912）

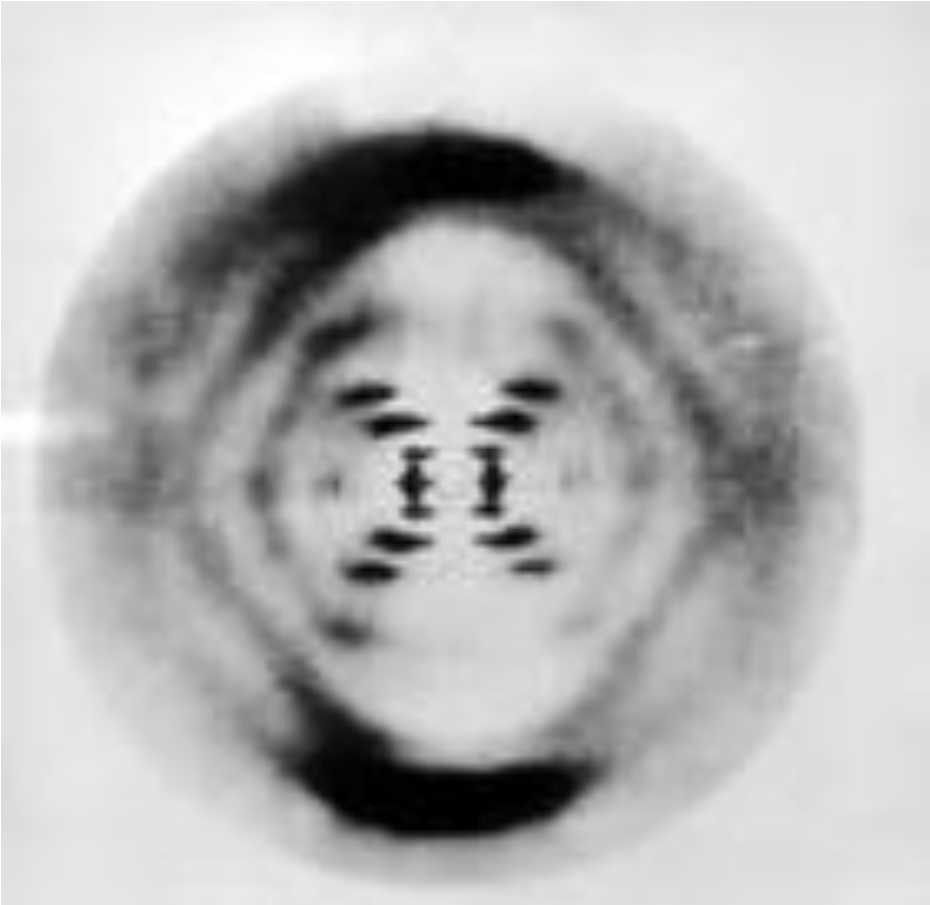
证实了*X*射线的波动性



证实晶格的周期性，反演晶格排布

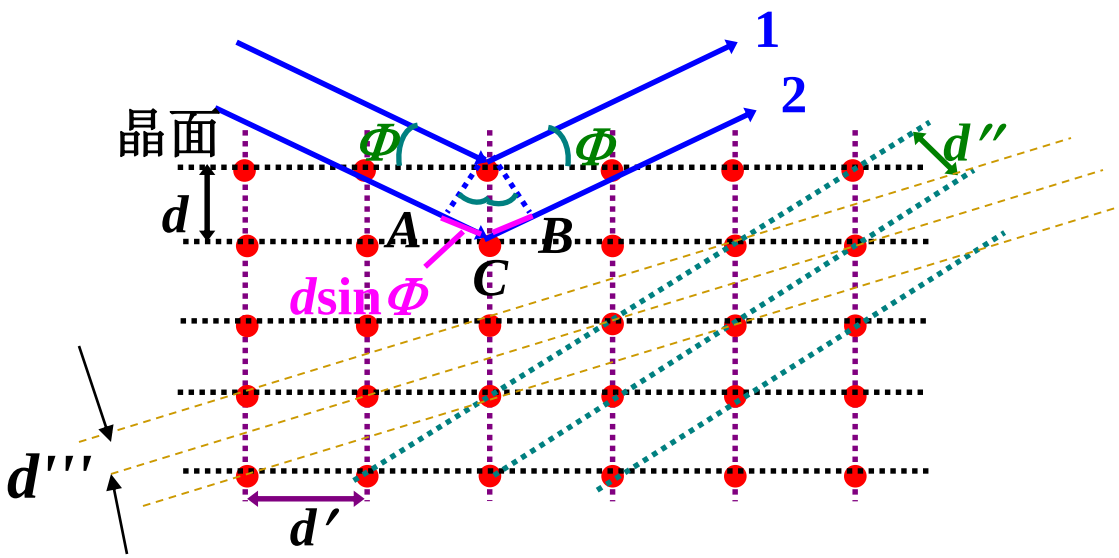


X-ray diffraction pattern for quartz (SiO_2).



Rosalind Elsie Franklin，1920－1958

二. X射线在晶体上的衍射



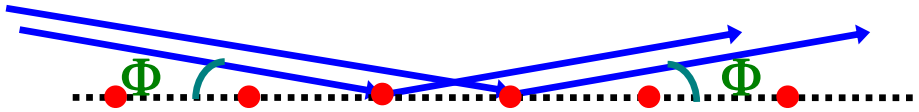
D ： 晶面间距（晶格常数）

Φ： 掠射角

1. 衍射中心

每个原子都是散射子波的子波源

2. 面上散射光的干涉



3 . 面间散射光的干涉

$$\delta = \overline{AC} + \overline{CB} = 2d \cdot \sin\Phi$$

散射光干涉加强条件：

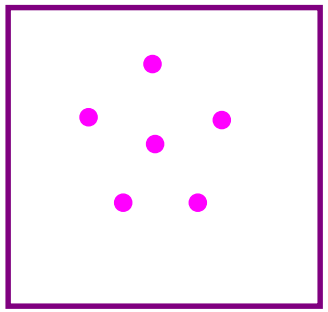
$$2d \cdot \sin\Phi = k\lambda \quad (k = 1, 2, \cdots)$$ —布喇格公式

三. 应用

- 已知Φ、λ 可测d — X射线晶体结构分析。
- 已知Φ、d可测λ — X射线光谱分析。

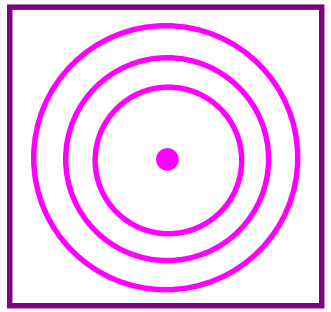
四. 实际观察X射线衍射的作法

1. 劳厄法：



劳厄相

2. 粉末法：



德拜相

单选题 1分

设置

波长为0.178nm的X射线以掠射角θ射向晶体表面时反射光出现第一级极大，已知晶格常数d= 0.178nm，则掠射角θ为

A 30°

B 45°

C 60°

D 75°

提交

（本题3分）波长为0.168 nm(1 nm = 10^{-9} m)的X射线以掠射角θ射向某晶体表面时，在反射方向出现第一级极大，已知晶体的晶格常数为0.168 nm，则θ角为[ ]

(A) 30°；

(B) 45°；

(C) 60°；

(D) 90°。